PHYSICS

1) The main scale of a vernier caliper reads in mm and its Vernier is divided into 10 divisions which coincide with 9 divisions of the main scale. The reading for the given situation will be :-

Fig 1

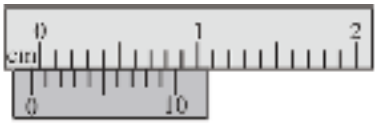
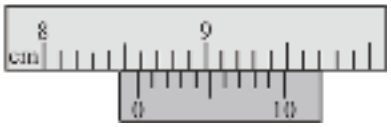


Fig 2



(1) 7.58 cm

(2) 8.79 cm

(3) 8.63 cm

(4) 9.78 cm

2) A false beam balance has unequal arm lengths. When an object placed on left pan it is balanced by 9 kg weight and when object is placed on right pan it is balanced by 16 kg weight. The true mass of object is :-

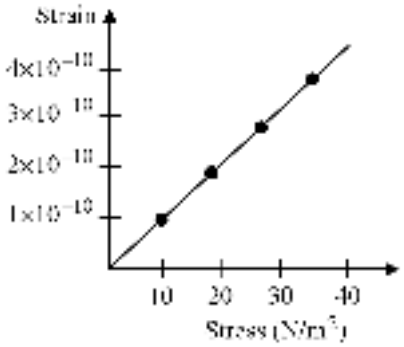
(1) 7.5 kg

(2) 9 kg

(3) 12 kg

(4) 16 kg

3) The stress-strain graph for a material is as shown in figure. The energy density in kJ/m^3 for a linear strain of 4.00×10^{-4} is :
(Assume material exhibits elastic behaviour upto linear strain of 4.00×10^{-4})



(1) 2 kJ/m^3

(2) 4 kJ/m^3

(3) 6 kJ/m^3

(4) 8 kJ/m^3

4) Two thin long glass tubes of diameter 2 mm and 5 mm are joined together with a common glass tube at base to form a U-tube with limbs as two capillaries. Now water is poured into U-tube, then difference in levels of water in two limbs is :
(Given : Surface tension of water = 7.2×10^{-2} Nm^{-1}, Angle of contact of water with glass = 0°, density of water = 10^3 kgm^{-3})

(1) 3.12 mm

(2) 4.32 mm

(3) 6.18 mm

(4) 8.64 mm

5) **Assertion (A) :-** In meter bridge experiment, a high resistance is always connected in series with galvanometer.
Reason (R) :- As resistance increases, current through circuit increases.
In light of above statements, choose the correct option :-

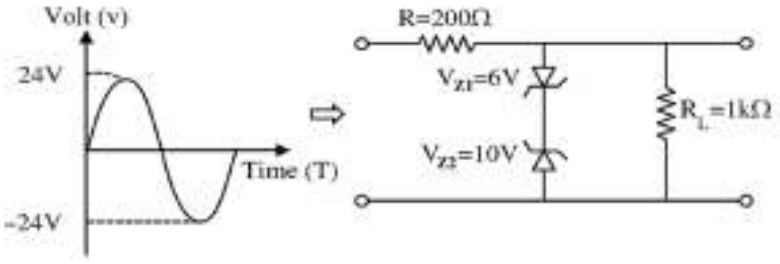
(1) Both (A) and (R) are correct and (R) is correct explanation of (A).

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

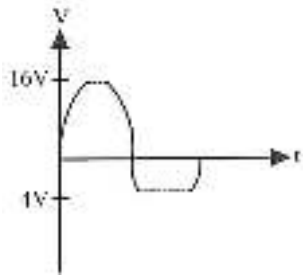
(3) (A) is correct but (R) is wrong.

(4) (A) is wrong but (R) is correct.

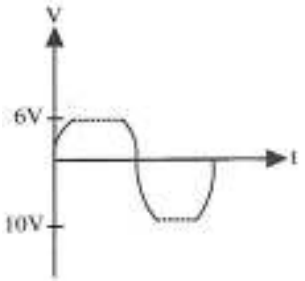
6) For the given circuit, applied waveform is shown near left of it. The correct output waveform will be : (Given : zener breakdown voltage V_{Z1} = 6V, V_{Z2} = 10V)

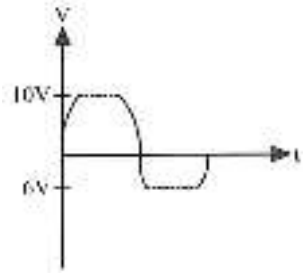


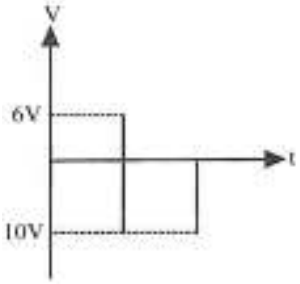
(1)

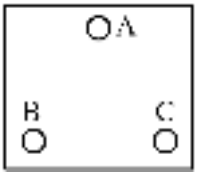


(2)



(3) 

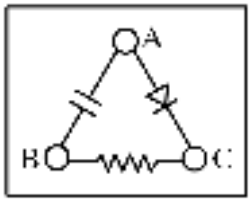
(4) 

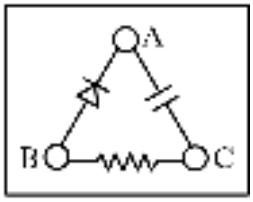
7) A student is given with a closed device with 3 terminals A, B & C outside. Inside the box, among any two terminals, three components resistor, capacitor and diode are connected. When box is connected with its any of two terminals with a variable dc supply of small emf some specific observations are obtained as listed below : 

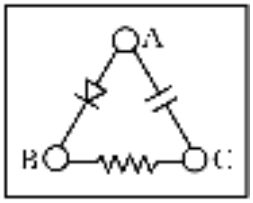
'+' ⇒ Positive polarity
'–' ⇒ Negative polarity

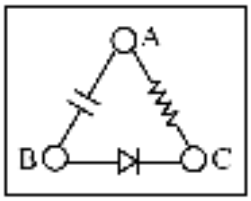
(i)	A^+, B^-	No current flows until about 0.7 V, after which current increases non linearly
(ii)	A^-, B^+	Very small saturation current
(iii)	B^+, C^-	Linear V-I characteristics with constant slope
(iv)	B^-, C^+	Same as in (iii)
(v)	A^+, C^-	Large initial current that rapidly decreases to zero
(vi)	A^-, C^+	Similar transient current as in (V)

The correct arrangement of components in box are :

(1) 

(2) 

(3) 

(4) 

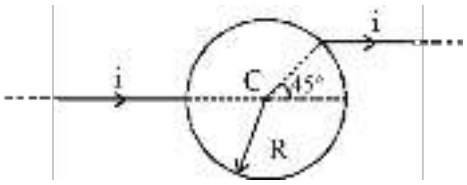
8) Force acting on a particle varies with distance travelled by a the particle as $F \propto \frac{1}{x^{1/3}}$, then power due to this force varies with x as:-

(1) $X^{1/3}$

(2) $X^{1/2}$

(3) $X^{-1/3}$

(4) X^{0}

9) Magnetic field at the centre C :- 

(1) $\frac{\mu_0 i}{4\pi R}\left(\sqrt{2}-1\right)$

(2) $\frac{\mu_0 i}{2\pi R}\left(\sqrt{2}-1\right)$

(3) $\frac{\mu_0 i}{2\pi R}\left(\frac{\sqrt{2}-1}{\sqrt{2}}\right)$

(4) Zero

10) A paramagnetic material has 10^{28} atom/m^3. Its magnetic susceptibility at temperature 350 K is 2.8×10^{-4}. Its susceptibility at 300 K is :-

(1) 3.672×10^{-4}

(2) 3.726×10^{-4}

(3) 3.267×10^{-4}

(4) 2.672×10^{-4}

11) A container has two chambers of volumes V_1 = 2 litres and V_2 = 3 litres separated by a partition made of a thermal insulator. The chambers contains n_1 = 5 and n_2 = 4 moles of ideal gas at pressures p_1 = 1 atm and p_2 =2 atm, respectively. When the partition is removed, the mixture attains an equilibrium pressure of :

(1) 1.3 atm

(2) 1.6 atm

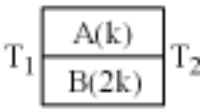
(3) 1.4 atm

(4) 1.8 atm

12) Two identical rods A and B of different materials are welded together as shown in figure. Their thermal conductivities are k and 2k. The ratio of equivalent thermal conductivity of the composite rods, when connected in series and in parallel connection?



(Fig-1)



(Fig-2)

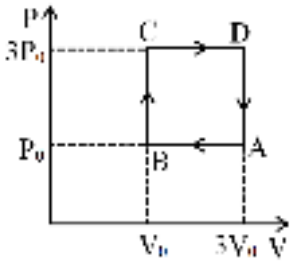
(1) $\frac{4}{3}$

(2) $\frac{3}{2}$

(3) $\frac{8}{9}$

(4) $\frac{9}{4}$

13) The P-V diagram represents the thermodynamic cycle of an engine, operating with an ideal monoatomic gas. The amount of heat extracted from the source and the efficiency in a single cycle is-



(1) $6P_0V_0$, 22.22%

(2) $10P_0V_0$, 33.33%

(3) $12P_0V_0$, 33.33%

(4) $18P_0V_0$, 22.22%

14) Two spheres of radii in the ratio 1 : 2 and densities in the ratio 2 : 1 and of same specific heat, are heated to same temperature and left in the same surrounding. Their rate of cooling will be in the ratio.

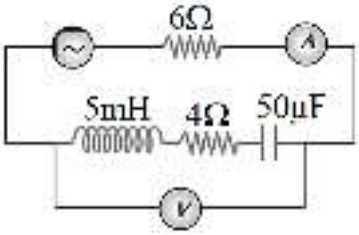
(1) 2 : 1

(2) 1 : 1

(3) 1 : 2

(4) 1 : 4

15) In the circuit shown in the figure, the ac source gives a voltage V= 20 cos(2000 t) volt. Neglecting source resistance, the voltmeter and ammeter reading will be :



(1) 0 V, 0.47 A

(2) 1.68 V, 0.47 A

(3) 0 V, 1.4 A

(4) 5.6 V, 1.4 A

16) A skylab of mass m kg is first launched from the surface of the earth in a circular orbit of radius 2R (from the centre of the earth) and then it is shifted from this circular orbit to another circular orbit of radius 3R. The minimum energy required to place the lab in the first orbit and to shift the lab from first orbit to the second orbit are

(1) $\frac{3}{4}mgR, \frac{mgR}{6}$

(2) $\frac{3}{4}mgR, \frac{mgR}{12}$

(3) mgR, mgR

(4) 2mgR, mgR

17) Potential energy of a particle at mean position is 4J and at extreme position it is 20J. Given that amplitude of oscillation is A. Match the following two columns.

	Column I		**Column II**
(a)	Potential energy at $x = \frac{A}{2}$	(P)	12 J
(b)	KE at $x = \frac{A}{4}$	(Q)	16 J
(c)	KE at x = 0	(R)	8 J
(d)	KE at $x = \frac{A}{2}$	(S)	None

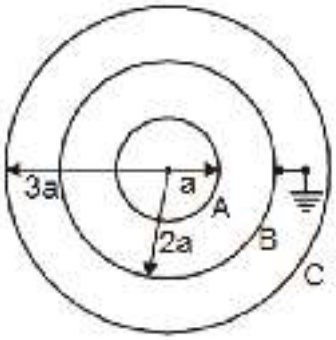
(1) a → R ; b → S ; c → Q ; d → P

(2) a → S ; b → P ; c → R ; d → Q

(3) a → P ; b → R ; c → P ; d → R

(4) a → Q ; b → Q ; c → S ; d → S

18) Figure shows a system of three concentric metal shells A, B and C with radii a, 2a and 3a respectively. Shell B is earthed and shell C is given a charge Q. Now if shell C is connected to shell



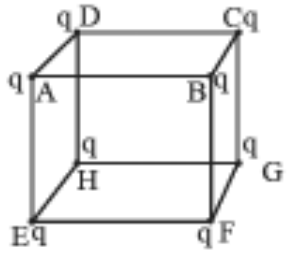
A, then the final charge on the shell B, is equal to :

(1) $-\frac{4Q}{13}$

(2) $-\frac{8Q}{11}$

(3) $-\frac{5Q}{3}$

(4) $-\frac{3Q}{7}$



19) Eight point charges each of charge q are fixed at vertices of a cube.

Statement : 1 electric flux through square surface ABCD of the cube is $\frac{q}{24\in_0}$

Statement : 2 electric flux through cube is $\frac{q}{\in_0}$

Statement : 3 electric flux through square surface EFGH of the cube is $\frac{q}{6\in_0}$

Which of the following statement are correct :

(1) 1 & 2

(2) 2 & 3

(3) 1 & 3

(4) All

20) A point particle of mass M is attached to one end of a massless rigid non-conducting rod of length L. Another point particle of same mass is attached to the other end of the rod. The two particles carry charges +q and & –q respectively. This arrangement is held in a region of uniform electric field E such that the rod makes a small angle θ (< 5°) with the field direction. The minimum time needed for the rod to become parallel to the field after it is set free.

(1) $2\pi\sqrt{\frac{ML}{2qE}}$

(2) $\pi\sqrt{\frac{ML}{2qE}}$

(3) $\frac{\pi}{2}\sqrt{\frac{ML}{2qE}}$

(4) $4\pi\sqrt{\frac{ML}{2qE}}$

21) A small square loop of side 'a' and one turn is placed inside a larger square loop of side b and one turn (b >> a). The two loops are coplanar with their centres coinciding. If a current I is passed in the square loop of side 'b', then the coefficient of mutual inductance between the two loops is :

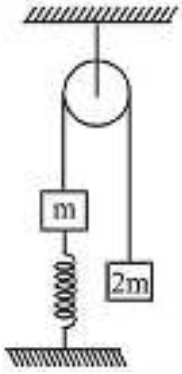
(1) $\frac{\mu_0}{4\pi}8\sqrt{2}\frac{a^2}{b}$

(2) $\frac{\mu_0}{4\pi}\frac{8\sqrt{2}}{a}$

(3) $\frac{\mu_0}{4\pi}8\sqrt{2}\frac{b^2}{a}$

(4) $\frac{\mu_0}{4\pi}\frac{8\sqrt{2}}{b}$

22) A block of mass 2m hangs by an inextensible light string that passes over a smooth pulley. Another mass m hangs from the other end of the string and is connected to a vertical massless spring. The system is stationary. Just after the string is cut, the accelerations of m and 2m, respectively, are :



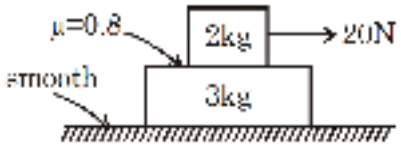
(1) 0, 0

(2) 0, g ↓

(3) $\frac{g}{2}$ ↓, g ↓

(4) 2g ↓, g ↓

23) Two blocks of mass 2kg and 3kg are placed over each other. Coefficient of friction between 2kg and 3kg blocks is μ = 0.8. Ground is perfectly smooth. Force of 20N acts on 2kg block as shown in figure. Select the **CORRECT** statement :-



(1) Friction force between 2kg and 3kg block is 8 N

(2) Friction force between 2kg and 3kg block is 16 N

(3) Acceleration of 3kg block is 16/3 m/s^2

(4) Acceleration of 3kg block is 4 m/s^2

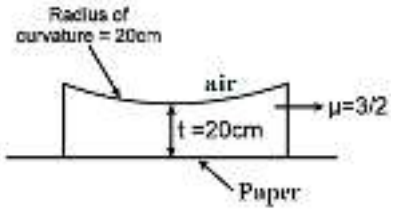
24) Two slits S_1 and S_2 illuminated by a white light source give a white central maxima. A transparent sheet of refractive index 1.25 and thickness t_1 is placed in front of S_1. Another transparent sheet of refractive index 1.50 and thickness t_2 is placed in front of S_2. If central maxima is not affected, then the ratio of $t_1 : t_2$ will be :-

(1) 1 : 2
(2) 2 : 1
(3) 1 : 4
(4) 4 : 1

25) If vibration frequency f of the star is given by the following equation : $f = KR^a \rho^b G^c$ where R is the radius of star, ρ is the density of star, G is the universal gravitational constant and K is a dimensionless constant, then (b + c) is equal to -

(1) 2
(2) 2.5
(3) 1
(4) 1.5

26) In the given figure a plano-concave lens is placed on a paper on which a flower is drawn. How far above its actual position does the flower appear to be?



(1) 10 cm
(2) 15 cm
(3) 50 cm
(4) None of these

27) A ray of monochromatic light is incident on one refracting face of a prism of angle 105°. It passes through the prism and is incident on the other face at the critical angle. If the refractive index of the material of the prism is $\sqrt{2}$, the angle of incidence on the first face of the prism is

(1) 30°
(2) 45°
(3) $\sin^{-1}\sqrt{3}$
(4) $\sin^{-1}\left(\sqrt{\frac{3}{2}}\right)$

28) In a compound microscope focal length of objective and eye piece are 2 cm and 3 cm respectively. If distance between both the lenses are 19 cm then find out magnifying power of compound microscope when final image formed at infinity.

(1) $\frac{475}{6}$

(2) $\frac{175}{3}$

(3) $\frac{225}{3}$

(4) None of these

29) If $\vec{A} = \hat{i} + 3\hat{j} - 2\hat{k}$ and $\vec{B} = 2\hat{i} - \hat{j} + 2\hat{k}$ than find a vector whose magnitude is equal to component of A along B and direction is along $\vec{A}$.

(1) $\frac{\hat{i} + 3\hat{j} - 2\hat{k}}{\sqrt{14}}$

(2) $\frac{+5(\hat{i} + 3\hat{j} - 2\hat{k})}{\sqrt{14}}$

(3) $\frac{5(-\hat{i} - 3\hat{j} + 2\hat{k})}{\sqrt{14}}$

(4) None of these

30) In a photoelectric effect experiment a light of frequency 1.5 times the threshold frequency is made to fall on the surface of photosensitive material. Now if the frequency is halved and intensity is doubled, the number of photo electrons emitted will be:

(1) Doubled

(2) Quadrupled

(3) Zero

(4) Halved

31) In Bohr's atomic model, the electron is assumed to revolve in a circular orbit of radius 0.5 Å. If the speed of electron is 2.2×10^6 m/s, then the current associated with the electron will be _____ $\times 10^{-2}$ mA. [Take π as $\frac{22}{7}$]

(1) 84

(2) 112

(3) 98

(4) 77

32)

On a planet particle is thrown with a velocity $\tilde{V} = 6\hat{i} + (20 - 4t)\hat{j}$ m/s. Then match the list.

List-I		List-II	
(P)	Time of flight (in seconds)	(1)	3
(Q)	Time when particle is moving at an angle of 53° with horizontal in upwards direction (in seconds).	(2)	10
(R)	Range of the particle (in meter)	(3)	50

(S)	Maximum height of the particle (in meter)	(4)	60

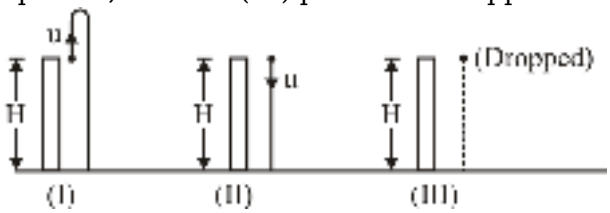
(1) P → 2; Q → 1; R → 4; S → 3

(2) P → 1; Q → 2; R → 3; S → 4

(3) P → 2; Q → 3; R → 4; S → 1

(4) P → 2; Q → 4; R → 1; S → 3

33) Three particles take times t_1, t_2 and t to reach ground from top of a tower in experiment I, II and III respectively. In (I) particles is projected up with speed u, in (II) particle is projected down with speed u, while in (III) particle is dropped. Then :



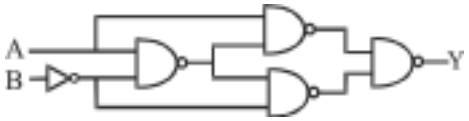
(1) $t = \sqrt{t_1 t_2}$

(2) $t = \frac{t_1 + t_2}{2}$

(3) $t = \frac{t_1 - t_2}{2}$

(4) None of these

34)



Output Y is :-

(1) $A\bar{B} + \bar{A}B$

(2) $\bar{A}\bar{B} + AB$

(3) $\frac{(A + B)}{\bar{A}}$

(4) $\frac{(A + B)}{\bar{B}}$

35) A Particle of mass M at rest decays into two particles of masses m_1 & m_2 having non zero Velocities. The ratio of the de-Broglie wavelength of the Particles λ_1/λ_2 is.

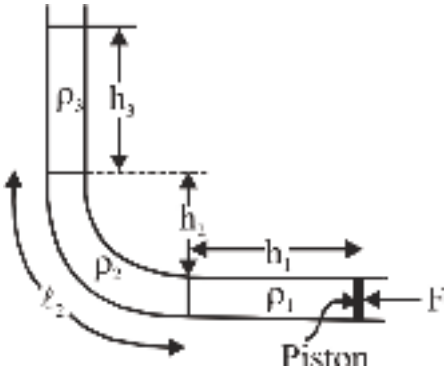
(1) m_1/m_2

(2) m_2/m_1

(3) 1

(4) $\sqrt{m_2}/\sqrt{m_1}$

36) A tube is bent into L shape and kept in a vertical plane. If these three liquids are kept in



equilibrium by the piston of area A, the value of $\frac{F}{A}$ is :

(1) $(\rho_1 h_1 + \rho_2 h_2 + \rho_3 h_3)g$

(2) $(\rho_1 h_1 + \rho_2 \square_2 + \rho_3 h_3)g$

(3) $(\rho_2 h_2 + \rho_3 h_3)g$

(4) $(\rho_2 \square_2 + \rho_3 h_3)g$

37) The magnetic field of an electromagnetic wave is given by
$\vec{B} = 1.6\times10^{-6}\cos(2\times10^{7}z + 6\times10^{15}t)(2\hat{i} + \hat{j})$ wb/m^2.
The associated electric field will be :-

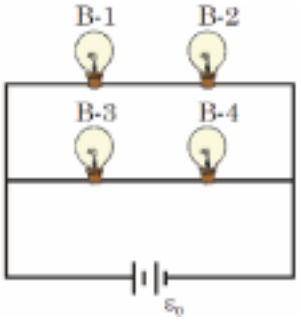
(1) $\vec{E} = 4.8 \times 10^{2} \cos(2 \times 10^{7} z + 6 \times 10^{15} t)\,(\hat{i} - 2\hat{j})$ V/m

(2) $\vec{E} = 4.8 \times 10^{2} \cos(2 \times 10^{7} z - 6 \times 10^{15} t)\,(2\hat{i} + \hat{j})$ V/m

(3) $\vec{E} = 4.8 \times 10^{2} \cos(2 \times 10^{-7} z - 6 \times 10^{15} t)\,(-2\hat{j} + \hat{i})$ V/m

(4) $\vec{E} = 4.8 \times 10^{2} \cos(2 \times 10^{+7} z + 6 \times 10^{15} t)\,(-\hat{i} + 2\hat{j})$ V/m

38) Four bulbs are connected into the circuit as shown in the circuit. The ratings of the bulbs are



shown below. The bulb which will be brightest

B-1 100V, 25W
B-2 50V, 50W
B-3 200V, 100W
B-4 50V, 25W

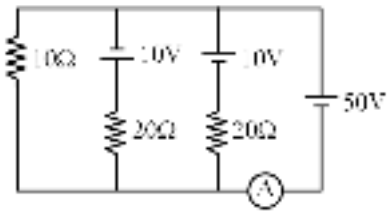
(1) B-1

(2) B-2

(3) B-3

(4) B-4

39) Reading of Ammeter will be :-



(1) 2 A

(2) 3 A

(3) 5 A

(4) 10 A

40) ***Assertion* (A) :-** When constant current is passing through a conductor of variable area of cross section, electric field inside conductor is inversely proportional to cross sectional area.

***Reason* (R) :-** Microscopic form of Ohm's law is $\vec{E} = \rho\vec{J}$, where $\vec{E}$ stands for electric field, ρ stands for resistivity, and $\vec{J}$ stands for current density.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is incorrect but **(R)** is correct

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

41) Find moment of inertia of a semi circular disc of mass M and radius R about axis passing through center of mass and perpendicular to its plane.

(1) $MR^2\left(\frac{1}{2}-\frac{16}{9\pi^2}\right)$

(2) $MR^2\left(\frac{1}{2}+\frac{16}{9\pi^2}\right)$

(3) $MR^2\left(\frac{1}{2}-\frac{9\pi^2}{16}\right)$

(4) $MR^2\left(\frac{1}{2}+\frac{9\pi^2}{16}\right)$

42) A circular platform is mounted on a frictionless vertical axle. Its radius R=2m and its moment of inertia about the axle is 200 kgm^2. it is initially at rest. A 50 kg man stands on the edge of the platform and begins to walk along the edge at the speed of 1ms^{-1} relative to the ground. Time taken by the man to complete one revolution is:

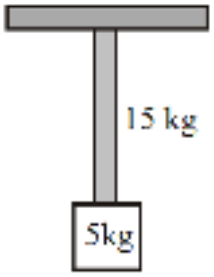
(1) π s

(2) $\frac{3\pi}{2}$s

(3) 2π s

(4) $\frac{\pi}{2}$s

43) A uniform rope of length 10 m and mass 15 kg hangs vertically from a rigid support. A block of mass 5 kg is attached to the free end of the rope. A transverse pulse of wavelength 0.08 m is produced at the lower end of the rope. The wavelength of the pulse when it reaches the top of the rope will be-



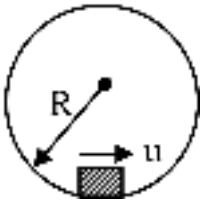
(1) 0.08 m

(2) 0.04 m

(3) 0.16 m

(4) 0 m

44) A particle is given an initial speed u inside a smooth spherical shell of radius R=1m such that it is just able to complete the circle. Acceleration of the particle, when its velocity is vertical is



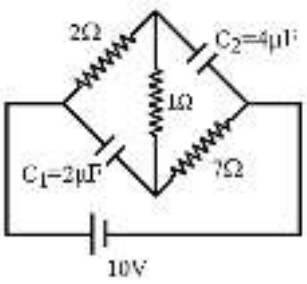
(1) $g\sqrt{10}$

(2) g

(3) $g\sqrt{2}$

(4) 3g

45) Find stored energy in 4 μF capacitor at steady state :



(1) 32 μ J

(2) 64 μ J

(3) zero

(4) 128 μ J

CHEMISTRY

1) In two H atoms A and B the electrons move around the nucleus in circular orbits of radius r and 4r respectively. The ratio of the time taken by them to complete one revolution is :-

(1) 1 : 4

(2) 1 : 2

(3) 1 : 8

(4) 2 : 1

2) If 'X' is ionization energy of hydrogen atom then the energy required for excitation of electron of Li^{2+} ion from 2nd excited state to 5th excited state is :-

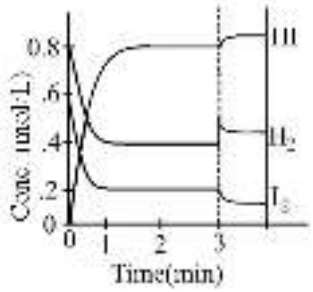
(1) $\frac{3X}{4}$

(2) $\frac{4}{3X}$

(3) $\frac{X}{12}$

(4) $\frac{12}{X}$

3) Variation of concentration with time for following reaction in the figure below is:



$H_2(g) + I_2(g)$ – heat ⇌ 2HI(g) At the instant 3 min, what change was imposed into the equilibrium ?

(1) Pressure was increased

(2) Temperature was decreased

(3) Temperature was increased

(4) Hydrogen was added

4)

The dissociation constant of NH_4OH is 2×10^{-5}. To obtain $[OH^-]$ of 1×10^{-6} M in 0.1M NH_4OH solution, the concentration of NH_4Cl in the solution would be :-

(1) 1.0 M

(2) 0.2 M

(3) 2.0 M

(4) 0.1 M

5) Match List-I with List-II

List-I		**List-II**	
(a)	1 mole water	(i)	22.4 L

(b)	1 g molecules of O_2 at NTP	(ii)	0.06 mole neutrons
(c)	12.022×10^{23} molecules of N_2H_4	(iii)	18 mL
(d)	0.16 g CH_4	(iv)	8 g atoms of H

(1) (a)–iii, (b)–i,(c)–iv, (d)–ii

(2) (a)–iv, (b)–i, (c)–iii, (d)–ii

(3) (a)–iii, (b)–iv, (c)–ii, (d)–i

(4) (a)–i, (b)–iii, (c)–ii, (d)–iv

6) A solution containing 2.7×10^{-3} mole of A^{2+} ions required 1.6×10^{-3} mole of MnO_4^- for the oxidation of A^{2+} to AO_3^- the medium use is :-

(1) Neutral

(2) Acidic

(3) Strong basic

(4) None of these

7) Calculate enthalpy change of the following reaction
$CH_2 = CH_2$ (g) + H_2 (g) → CH_3–CH_3 (g)
The bond energy of C–H, C–C, C = C, H–H are 414, 347, 615 and 435 kJ mol^{-1} respectively:-

(1) –250 kJ

(2) +250 kJ

(3) +125 kJ

(4) –125 kJ

8) 10 ml each of 1M HCl and 1M H_2SO_4 are completely neutralised by 1M NaOH solution that liberate the heat of x and y respectively. Which is true :-

(1) 2x = y

(2) x = y

(3) x = 2y

(4) 3x = y

9) If a gas absorbs 200 J of heat and expands by 500 cm^3 against a constant pressure 2×10^5 Nm^{-2} then change in internal energy is :-

(1) – 300 J

(2) – 100 J

(3) + 100 J

(4) + 300 J

10) The Henry's law constant (K_H) values of three gases (A, B, C) in water are 145, 2×10^{-5} and 35

kbar, respectively. The solubility of these gases in water follow the order :

(1) B > A > C

(2) B > C > A

(3) A > C > B

(4) A > B > C

11) Given below are two statements
Statement I : Mixture of Chloroform and Acetone shows negative deviation from Raoult's law.
Statement II : For ideal solution $(\Delta G)_{mixing} = 0$.
In the light of above statements, choose the most appropriate answer from the options given below.

(1) Both statement I and statement II are incorrect.

(2) Statement I is correct but statement II is incorrect

(3) Statement I is incorrect but statement II is correct

(4) Both statement I and statement II are correct

12) Find the equilibrium constant at 298 K for the reaction
$Cu^{+2}(aq) + In^{+2}(aq) \rightleftharpoons Cu^{+}(aq) + In^{+3}(aq)$
Given $E^{o}_{Cu^{+2}/Cu^{+}} = 0.15\ V$
$E^{o}_{In^{+3}/In^{+}} = -0.42\ V, E^{o}_{In^{+2}/In^{+}} = -0.40\ V$

(1) 10^{20}

(2) 10^{10}

(3) 10^{5}

(4) None of these

13) How much will the oxidation potential of a hydrogen electrode change when its solution initially at pH = 0 is neutralised to pH = 7 ?

(1) increases by 0.059 V

(2) decreases by 0.059 V

(3) increases by 0.413 V

(4) decreases by 0.413 V

14) The reaction : $2A \rightarrow 2B + C$ occurs by the mechanism :
$A \xrightarrow{k_1} B + X$ [slow] ...(1)
$A + X \xrightarrow{k_2} B + C$ [fast] ...(2)
Which of the following statement is incorrect ?

(1) Molecularity of the reaction is meaningless

(2) The order of the reaction with respect to A is two

(3) The rate law of reaction is : $r = k_1[A]$

(4) $k_2 > k_1$

15) The rate law for a reaction A + B → product is $r = K[A]^2[B]^{-1}$ then

which of the following statement is false :-

(1) If [B] is held constant while [A] is doubled, the rate of reaction will be quadrupled
(2) If [A] is held constant while [B] is reduced to one quarter, the rate will be quadrupled
(3) If [A] & [B] are both doubled, the rate of reaction will be quadrupled
(4) This is a first order reaction

16) Which of the following given pair of radius is correct?

(1) Li < I and $Li^+ < I^-$
(2) Li < Mg and $Li^+ < Mg^{+2}$
(3) Na > Ca and $Na^+ > Ca^{+2}$
(4) K > Sr and $K^+ > Sr^{+2}$

17) Select the correct order of electronegativity.

(1) B > Al < Ga < In < Tl
(2) F > Cl < Br < I
(3) P > O > S > F
(4) All are correct.

18) Which of the following statement is correct :

(1) I_2Cl_6 is nonpolar compound
(2) CH_3F is more polar than CH_3Cl
(3) CO_2 cannot act as Lewis acid
(4) Urea has intramolecular H-bond

19) Match the column -

Compound		**Properties**	
(i)	C_2H_4	(P)	p – p overlapping
(ii)	C_2H_2	(Q)	sp – sp overlapping
(iii)	$CH_2 = C = CH_2$	(R)	sp^2 – s overlapping
(iv)	C_2H_6	(S)	lone pairs Zero

(1) (i)-P,R ; (ii)-P,Q,R ; (iii)-P,R ; (iv)-S
(2) (i)-P,R,S ; (ii)-P,Q,S ; (iii)-P,R,S ; (iv)-S
(3) (i)-R,S ; (ii)-P,Q ; (iii)-P ; (iv)-P,S
(4) (i)-Q,R,S ; (ii)-P,R ; (iii)-Q,R ; (iv)-R,S

20) Correct relation is ?

(Where B = Bond pair, E = Lone pair)

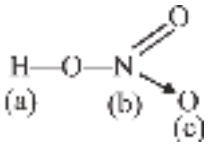
(1) AB_3E_2 → See-Saw shape

(2) AB_4E → Folded square

(3) AB_2E_2 → Angular

(4) Both (2) & (3)

21) For the following molecule calculate formal charge on a,b & c respectively ?



(1) 0, –1, +1

(2) +1, 0, –1

(3) –1, 0, +1

(4) 0, +1, –1

22) Which of the following complex is colourless ?

(1) $Na_2[Fe(CN)_5(NO)]$

(2) $[Cu(NH_3)_4]SO_4$

(3) $K_3[Cu(CN)_4]$

(4) $[Ti(H_2O)_6]Cl_3$

23) Which of following is not considered as organometallic compound ?

(1) Ferrocene

(2) Ziese's salt

(3) Grignard reagent

(4) Cis-platin

24) Match the Column -

Column - I		**Column - II**	
(A)	Oxalato	**(i)**	Denticity = 3 Charge = 0
(B)	Ethylenediamine triacetato	**(ii)**	Denticity = 1 Charge = 0
(C)	Dimethylglyoximato	**(iii)**	Denticity = 2 Charge = –2
(D)	triphenylphosphine	**(iv)**	Denticity = 2 Charge = –1
(E)	Diethylene triamine	**(v)**	Denticity = 5 Charge = –3

(1) (A)-(iv); (B)-(i); (C)-(v); (D)-(ii); (E)-(iii)

(2) (A)-(iv); (B)-(ii); (C)-(iii); (D)-(i); (E)-(v)

(3) (A)-(iii); (B)-(v); (C)-(iv); (D)-(ii); (E)-(i)

(4) (A)-(iii); (B)-(v); (C)-(ii); (D)-(iv); (E)-(i)

25) The correct statement among the following regarding group 16 elements is

(1) All elements exhibit allotropy.

(2) The large difference between the melting point and boiling point of oxygen and sulphur may be explained on the basis of their atomicity.

(3) The stability of –2 oxidation state decreases down the group.

(4) All of the above

26)

The incorrect statements is :

(1) $SnCl_4$ is more covalent than $SnCl_2$

(2) PbI_4 commonly exist at room temperature

(3) GeX_4 is more stable than GeX_2

(4) PbF_4 exist at room temperature

27) What is X and Y in the given reactions?
$2X_2(g) + 2H_2O(\square) \rightarrow 4HX\ (aq) + O_2\ (g)$
$Y_2\ (g) + H_2O\ (\square) \rightarrow HY\ (aq) + HOY(aq)$

(1) X = Cl, Y = F

(2) X = Cl, Y = Br

(3) X = F, Y = Cl

(4) X = I, Y = F

28) Select the incorrect set with colour :-

(1) $Fe^{+2}_{(aq.)}$ - Green

(2) $KMnO_4$ - almost black

(3) ZnS - Yellow

(4) PbS → Black

29) A lanthanoid has 'n' electrons in 4f subshell and another lanthanoid has (14 – n) electrons in 4f subshell. It means:-

(1) Both lanthanoid shows similar oxidising power

(2) Both shows same magnetic behaviour

(3) Both (1) and (2)

(4) Both are radioactive

30) When solution containing CrO_4^{2-} ion is mixed with Ba^{+2}, Sr^{+2} and Ca^{+2} solution. Whose precipitate

will be obtained first ?

(1) $CaCrO_4$

(2) $SrCrO_4$

(3) $BaCrO_4$

(4) Mixture of all

31) 1.216 gm of an organic compound undergoes to kjeldahl's method, the ammonia evolved was absorbed in 100 ml 1N H_2SO_4, the residual acid solution was made upto 500 ml by addition of water 20 ml of this solution required 32 ml of N/10 caustic soda solution for complete neutralisation, the percentage of nitrogen in organic compound.

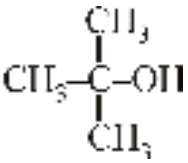
(1) 46%

(2) 56%

(3) 23%

(4) 60%

32)

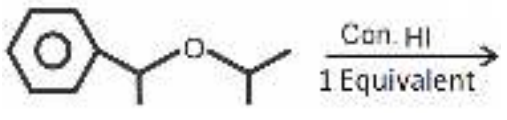
Column-I (Compound)		**Column-II (Test)**	
(a)	CH_3-CH_2-OH	(p)	It gives compound having unpleasant odour while treating with $CHCl_3$ & alc. KOH
(b)	$CH_3-CH_2-NH_2$	(q)	It gives yellow ppt while treating with I_2 & NaOH
(c)		(r)	It reacts readily with Lucas reagent.
(d)		(s)	It gives white ppt while treating with Br_2/H_2O

(1) a → q, b → p, c→ s, d → r

(2) a → q, b → p, c→ r, d → s

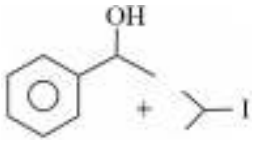
(3) a → p, b → q, c→ s, d → r

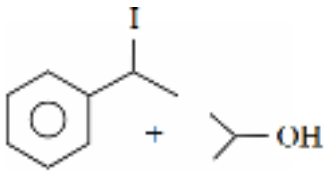
(4) a → s, b → r, c→ q, d → p

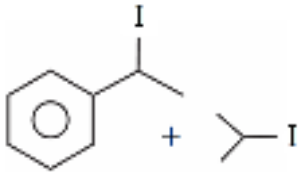
33) 

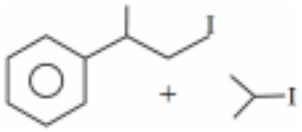
P + Q

P and Q are –

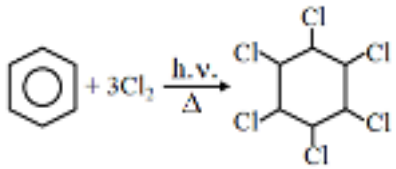
(1) 

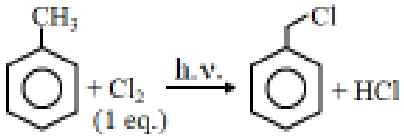
(2) 

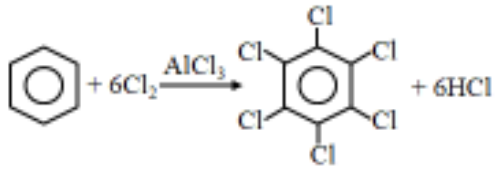
(3) 

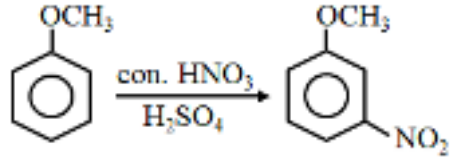
(4) 

34) Which reaction show incorrect product-

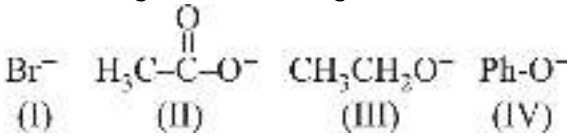
(1) 

(2) 

(3) 

(4) 

35) Arrange the following in order of their leaving group tendency?



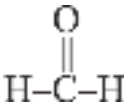
(1) I > II > III > IV

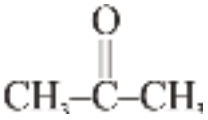
(2) I > III > II > IV

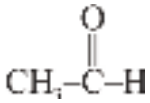
(3) I > II > IV > III

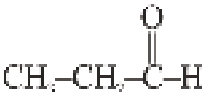
(4) I > IV > III > II

36) Compare reactivity order towards NAR ?

(P) 

(Q) 

(R) 

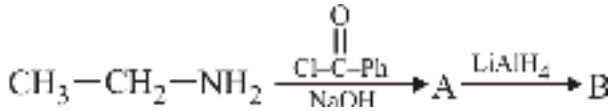
(S) 

(1) P > Q > R > S

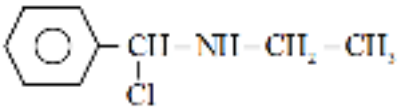
(2) P > R > S > Q

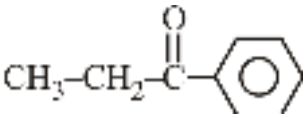
(3) P > R > Q > S

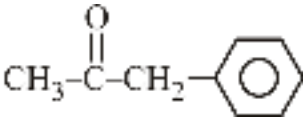
(4) Q > S > R > P

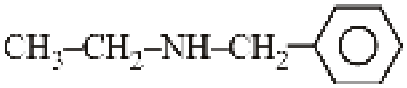
37) 

B is:-

(1) 

(2) 

(3) 

(4) 

38)

Given below are two statements :

Statement I : Benzenediazonium salt is prepared by the reaction of aniline with nitrous acid at 273-278 K. It decomposes easily in the dry state.

Statement II : Insertion of iodine into the benzene ring is difficult and hence iodobenzene is prepared through the reaction of benzenediazonium salt with KI.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(1) Both Statement I and Statement II are correct

(2) Both Statement I and Statement II are incorrect

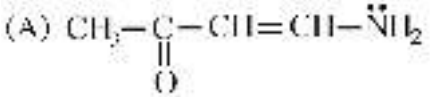
(3) Statement I is correct but Statement II is incorrect

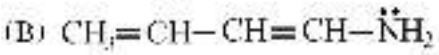
(4) Statement I is incorrect but Statement II is correct

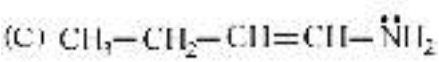
39) Which of the following carbohydrates does not show mutarotation :

(1) Glucose
(2) Fructose
(3) Sucrose
(4) Maltose

40) Compare the C–N bond length in following compounds.

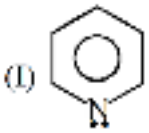
(A) 

(B) 

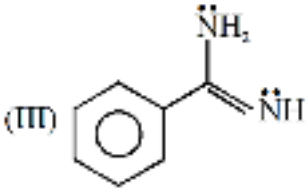
(C) 

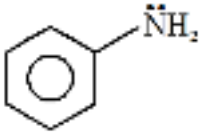
(1) A > B > C
(2) B > A > C
(3) A > C > B
(4) C > B > A

41) The decreasing order of basic characters of the following Nitrogen is :

(I) 

(II) 

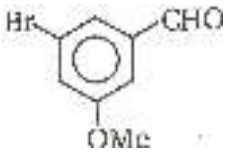
(III) 

(IV) 

(1) III > I > IV > II
(2) II > I > IV > III
(3) IV > III > II > I
(4) III > I > II > IV

42) Arrange the following in their boiling points :
(i) n-butane
(ii) iso-butane
(iii) n-pentane
(iv) iso-pentane
(v) neopentane

(1) iii > i > ii > iv > v
(2) v > iv > ii > i > iii
(3) iii > iv > v > i > ii
(4) ii > i > v > iv > iii

43) The correct IUPAC name of compound is 

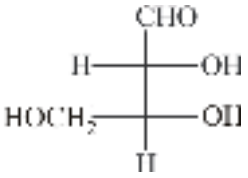
(1) 3-Methoxy-5-bromo benzene carbaldehyde

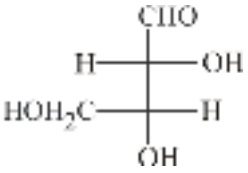
(2) 3-Formyl-5-bromo phenyl methyl ether

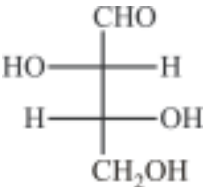
(3) 3-Bromo-5-methoxy benzaldehyde

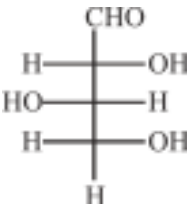
(4) 3-Formyl-5-bromo-1-methoxy benzene

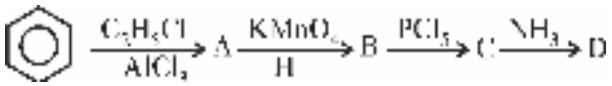
44) Which of the following is *D-Erythrose* ?

(1) 

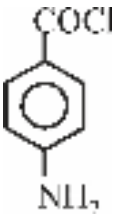
(2) 

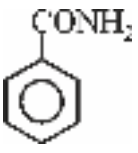
(3) 

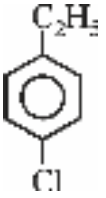
(4) 

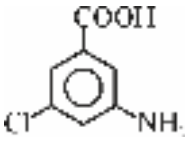
45) 

D is :

(1) 

(2) 

(3) 

(4) 

BIOLOGY

1) In homosporous pteridophyte, the gametophytes is :-

(1) vascular

(2) monoecious

(3) dioecious

(4) dependent on sporophyte

2) Study the following statements and choose the correct option -
(a) *Solanum, Petunia* and *Datura* are different genera but placed in same family.
(b) The sex organs are absent and the asexual spores are generally not found in both basidiomycetes and deuteromycetes
(c) Bryophytes have great ecological importance but few economical importance
(d) *Selaginella* and *Equisetum* form distinct compact structure are called strobili or cones
(e) Unbranched stem present in *Cedrus* while branched stem present in *Pinus*

(1) All are correct except 'b'

(2) Only b and c correct

(3) Only d and e correct

(4) All are incorrect except a, c & d

3) Read the following example :-
Chara, *Ectocarpus*, *Spirogyra*, *Volvox*, *Polysiphonia*, *Porphyra*
How many examples have chlorophyll 'a' and 'b'?

(1) Three

(2) Four

(3) Five

(4) Six

4) Match the column-I and Column-II and select correct option :-

Column-I		**Column-II**	
A	R.H. Whittaker	i	Carageen
B	Archaebacteria	ii	Production of methane
C	Red algae	iii	Fucoxanthin
D	Brown algae	iv	Five kingdom system of classification

(1) A → iii, B → i, C → iv, D → ii

(2) A → i, B → ii, C → iii, D → iv

(3) A → ii, B → i, C → iv, D → iii

(4) A → iv, B → ii, C → i, D → iii

5) Identify 'A' in given figure – 

(1) Anthenidiophore

(2) Archegoniophore

(3) Capsule

(4) Leaves

6) Androecium of Papilionaceae and Malvaceae family respectively is :

(1) Monodelphous & Diadelphous

(2) Diadelphous and Monodelphous

(3) Diadelphous and Polyadelphous

(4) Monodelphous & Polyadelphous

7) **Statement-I :** In monocot plants, the primary root is short lived and replaced by a large number of roots.
Statement-II : Stem is covered at the apex by thimble like structure which protects the tender apex of the stem.

(1) Statement-I is true and Statement-II is false

(2) Statement-I is false and Statement-II is true

(3) Statement-I and Statement-II both are correct

(4) Statement-I and Statement-II both are incorrect

8) Match the following.

	Column-I		**Column-II**
A.	Hypodermis in dicot stem	(i)	Absent
B.	Pericycle in dicot stem	(ii)	Parenchymatous
C.	Ground tissue in monocot stem	(iii)	Collenchymatous
D.	Phloem parenchyma in monocot stem	(iv)	Sclerenchymatous

(1) A-(iii), B-(iv), C-(ii), D-(i)

(2) A-(i), B-(ii), C-(iv), D-(iii)
(3) A-(iv), B-(i), C-(iii), D-(ii)
(4) A-(ii), B-(iii), C-(i), D-(iv)

9) The distinguishing feature of open vascular bundles is presence of-

(1) Endodermis
(2) Xylem and phloem
(3) Pericycle
(4) Cambium

10) Crossing over during meiosis increases genetic variation. During which substage does it occur?

(1) Leptotene
(2) Pachytene
(3) Diplotene
(4) Diakinesis

11) A chromosome with a very short arm and a very long arm is referred to as

(1) Metacentric
(2) Telocentric
(3) Acrocentric
(4) Sub-metacentric

12) Choose the correct statements :
(A) RER is frequently observed in the cells actively involved in lipid synthesis and secretion.
(B) The number of mitochondria per cell is variable depending on the physiological activity of the cells.
(C) In the chromoplasts fat soluble carotenoid pigments like carotene, xanthophylls & other are present.
(D) Cilia are small structures which work like oars, causing the movement of either the cell or the surrounding fluid.

(1) A, B, C
(2) B, C, D
(3) A, C, D
(4) A, B, D

13) If a cell has 46 chromatids in metaphase, then how many chromatids are present in anaphase for that cell?

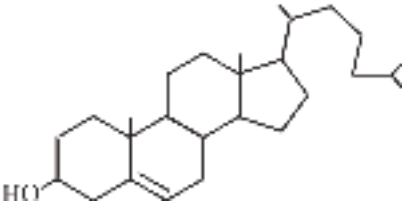
(1) 92
(2) 46
(3) 23
(4) 69

14) Select the correct statements.
I. The detailed structure of the membrane was studied only after the advent of the electron microscope in the 1950s.
II. Nonpolar tail of phospholipid is made up of saturated hydrocarbon.
III. The ratio of lipid and protein are constant in different cell types.
IV. In human beings, the membrane of the erythrocyte has approximately 52% lipid and 40% protein.
V. Depending upon the ease of extraction, membrane proteins can be classified as integral and peripheral proteins.

(1) I, II, III, IV
(2) I and II only
(3) III, IV, V
(4) I, II, V

15) Which one of the following enzymes contains 'Haem' as the prosthetic group ?

(1) RuBisCo
(2) Carbonic anhydrase
(3) Succinate dehydrogenase
(4) Catalase

16) Given structural formula is correctly identified along with its related function by which of the following options? 

(1) Cholesterol – A component of animal cell membrane
(2) Lecithin – A component of cell membrane
(3) Triglyceride – An energy source
(4) Cholesterol – A simple lipid.

17) Match the column-I with column-II and choose the correct option :-

	Column-I		**Column-II**
(A)	RuBisCO	(i)	Hormone
(B)	Collagen	(ii)	Enzyme
(C)	Trypsin	(iii)	Most abundant protein in the animal world
(D)	Insulin	(iv)	Most abundant protein in whole of the biosphere

(1) A-i, B-ii, C-iii, D-iv
(2) A-iv, B-iii, C-i, D-ii
(3) A-iv, B-iii, C-ii, D-i

(4) A-iii, B-iv, C-ii, D-i

18) Match the column I, II and III and identify the correct match :-

	Column-I		**Column-II**		**Column-III**
A	Auxin	P	GA	i	Fruit ripening
B	Gibberellins	Q	Kinetin	ii	Nutrient mobilisation
C	Ethylene	R	IAA	iii	Malting producess
D	Cytokinin	S	Ethephon	iv	Xylem differentiation

(1) A-P-i, B-Q-ii, C-R-iii, D-S-iv

(2) A-R-iv, B-P-iii, C-S-i, D-Q-ii

(3) A- S-iv, B-R-iii, C-Q-ii, D-P-i

(4) A-Q-i B-P-ii, C-S-iii, D-R-iv

19) Which of the following statement is incorrect regarding C-4 pathway :-

(1) Oxaloacetate is first stable product of C-4 cycle

(2) PEP carboxylase is CO_2 fixing enzyme in mesophyll cell

(3) Maize and sorghum are C-4 plant

(4) C-4 pathway is also called Calvin cycle

20) For the synthesis of 2 molecules of glucose through Calvin cycles, how many ATP molecules are required?

(1) 2

(2) 6

(3) 18

(4) 36

21) **Assertion :-** There is very high transport of gases from one plant part to another
Reason :- Each plant part takes care of its own gas-exchange needs.

(1) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

(2) Assertion is true but Reason is false.

(3) Assertion is false but Reason is true.

(4) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

22) The complex II of mitochondrial electron transport chain is also known as

(1) Cytochrome bc_1

(2) Succinate dehydrogenase

(3) Cytochrome c oxidase

(4) NADH dehydrogenase

23) The first step of EMP pathway in obligate aerobic organisms involves–

(1) Phosphorylation of glucose in the presence of oxygen with the help of an enzyme isomearse

(2) No use of ATP molecules as energy source

(3) Formation of glucose-6-phosphate in presence of O_2 only

(4) Phosphorylation of glucose using ATP in presence of an enzyme hexokinase and no oxygen is required

24) The characteristic features of angiospermic microsporangium
a. It is generally surrounded by 4 wall layers.
b. Outer 3 wall layers that involve in the function of protection and also help in dehiscence of anther to release the pollen.
c. The inner most wall layer that helps in nourishes the developing pollen grains-endothecium
d. Cell of tapetum are multinucleated.

(1) a, c & d are correct

(2) c & d are correct

(3) a, b & d are correct

(4) a ,b, c, d correct

25) **Assertion (A):** In a majority of flowering plants monosporic development occurs.
Reason(R): Out of four megaspores form from one MMC only one is functional while the other three degenerate and functional megaspore develops into the female gametophyte.

(1) Assertion (A) and Reason (R) are true, and R is the correct explanation of A.

(2) Assertion (A) and Reason (R) are true, but R is not the correct explanation of A.

(3) Assertion (A) is true, but Reason (R) is false

(4) Assertion (A) is false, but Reason (R) is true.

26) The portion of the embryonal axis above the level of attachment of cotyledon is -

(1) Hypocotyl

(2) Epicotyl

(3) Radicle

(4) Coleorhiza

27) The Punnett square shown below represents the pattern of inheritance in dihybrid cross -

	RY	Ry	rY	ry
RY	F	J	N	R
Ry	G	K	O	S
rY	H	L	P	T
ry	I	M	Q	U

Choose the correct symbol having similar genotype

(1) F, K

(2) G, H

(3) L, O

(4) P, U

28) What is the pattern of inheritance for polygenic trait?

(1) Mendelian inheritance pattern

(2) Non-mendelian inheritance pattern

(3) Autosomal dominant pattern

(4) X-linked recessive inheritance pattern

29) Read the following statements and choose the set of correct option.
(i) Henking observed in a some member of insects, the mechanism of sex determination is of the XO type.
(ii) Female grasshopper has a pair of X-chromosomes but male grasshopper has only one Y-chromosomes.
(iii) In human beings and in Drosophila the male have one X and one Y chromosome besides autosomes.
(iv) Male grasshopper show heterogamety.

(1) (i), (ii), (iii)

(2) only (ii), (iii)

(3) (iii), (iv)

(4) only (iii)

30) Which of the following is not a correct statement.

(1) In prokaryotes the DNA in nucleoid is organised in large loops held by proteins.

(2) DNA has evolved from RNA with chemical modifications

(3) During transcription RNA polymerase binds to start codon AUG.

(4) The human genome contains 3164.7 million bp.

31) Read the following statements and choose the incorrect statements :
(a) Deoxyribonucleoside triphosphates serve dual purposes in DNA replication.
(b) The coding sequences or expressed sequences are defined as introns.
(c) The split-gene arrangements represents probably modern feature of the genome.
(d) The codon is read in mRNA in a discontinuous fashion.
(e) Untranslated regions are required for efficient translation process.
Choose the correct option :

(1) a, b and c

(2) b, c and d

(3) c, d and e

(4) a, d and e

32) Frame shift mutation is the genetic basis of proof that codon is-

(1) Triplet and commaless

(2) Universal

(3) Degenerate

(4) Non universal

33) Match List-I with List-II.

List-I		List-II	
(A)	RNA polymerase + σ – factor	I.	Ribozyme
(B)	RNA polymerase + ρ – factor	II.	Catalyses the process of elongation
(C)	RNA Polymerase	III.	Initiate the process of transcription
(D)	23 S rRNA	IV	Terminate the process of transcription

(1) A-I, B-II, C-IV, D-III

(2) A-III, B-IV, C-I, D-II

(3) A-III, B-IV, C-II, D-I

(4) A-II, B-IV, C-I, D-III

34) Given below are two statement : one is labelled as Assertion A and other is labelled as Reason R
Assertion :- One of the H.G.P. methodology involves sequencing the whole set of genome that contained all the coding and non coding sequences and later assigning different regions in the sequence with functions.
Reason :- Repetitive sequences are stretches of DNA sequences that are repeated many times, having no direct coding function but they shed light on chromosome structure, dynamics and evolution.
In the light of above statements, choose correct answer from the options given below :-

(1) A and R both are true and R is the correct explanation of A.

(2) A is true but R is false

(3) A is false but R is true

(4) A and R both are true but R is not correct explanation of A

35) Read the following statements and choose the correct set of statements:
(a) The Human Genome Project was a 13-year project coordinated by the U.S. Department of Energy and the National Institute of Health.
(b) The codon is read in mRNA in a contiguous fashion
(c) If an inheritable mutation is observed in a population at very low frequency, it is referred to as DNA polymorphism
(d) Insertion or deletion of one or two bases changes the reading frame from the point of insertion or deletion
Choose the correct answer from the options given below :-

(1) (a),(b) and (d) only

(2) (c) and (d) only

(3) (a),(b),(c) and (d) only

(4) (a),and (b) only

36)

Given below are two statements :
Statement-I : Failure of cytokinesis after telophase stage of cell division results in an increase in a whole set of chromosome in an organism and this phenomenone is known as polyploidy.
Statement-II : Failure of segregation of chromatids during cell division cycle results in the gain or loss of chromosome(s), called aneuploidy.
In the light of the above statements choose the correct answer from the options given below :

(1) **Statement I** is true but **Statement II** is false.

(2) **Statement I** is false but **Statement II** is true

(3) Both **Statement I** and **Statement II** are true.

(4) Both **Statement I** and **Statement II** are false.

37) Select the incorrect statement :-

(1) Polymorphism arises due to mutations.

(2) Polymorphism does not play any role in evolution.

(3) The VNTR belongs to a class of satellite DNA referred to as mini-satellite.

(4) The size of VNTR varies in size from 0.1 to 20 kb.

38) Potent force in organic evolution is :-

(1) Predation

(2) Interspecific competion

(3) Exploitation

(4) Parasitism

39) **Statement I :** Predators acting as conduits for energy transfer across trophic levels.
Statement II : The prickly pear cactus introduced into Australia in the early 1920's caused havoc by spreading rapidly into millions of hectares of rangeland.
In the light of above statements, choose the most appropriate answer from the option given below :

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

40) Fill in the blank:
In terrestrial ecosystem pyramid of biomass is generally _____ In an aquatic ecosystem it is often _____ due to rapid turn over of phytoplanktons and due to their shorter life span.

(1) Upright; Upright

(2) Upright; Inverted

(3) Inverted; Upright

(4) Inverted; Inverted

41) Standing crop is a measure of:-

(1) Biomass

(2) Mineral nutrients in soil

(3) Sunlight energy

(4) Amount of energy available for consumption of next trophic level

42) Fill in the blanks -
__A__ break down detritus into smaller particles. This process is called __B__ by the process of __C__ water soluble nutrients go down into the soil horizon.

(1) A–Detritivores B–Fragmentation C– Leaching

(2) A–Detritivores B–Humification C–Catabolism

(3) A–Decomposers B–Leaching C–Humification

(4) A–Decomposers B–Leaching C– Catabolism

43) Choose the correct option of percentage of ascending order of given living beings species which faces threat of extinction at presently-
Birds, Mammals, Gymnosperms, Amphibians

(1) Amphibians, Gymnosperms, Mammals, Birds

(2) Birds, Mammals, Gymnosperms, Amphibians

(3) Birds, Amphibians, Mammals, Gymnosperms

(4) Mammals, Birds, Gymnosperms, Amphibians

44) The value of Z for frugivorous birds in tropical forest of different continents is

(1) 0.1 – 0.2

(2) 0.6 – 1.2

(3) 1.15

(4) 2.30

45) Which of the following is correctly matched:
(A) Carrot grass – Lantana
(B) Farmers friend – fungi
(C) Amazon rain forest – Lungs of planet
(D) World summit – Johannesburg

(1) Only A,B

(2) C,D

(3) Only D

(4) A,B,D

46) Animal that usually undergoes external fertilisation with indirect development is–

(1) *Hyla*
(2) *Scoliodon*
(3) *Taenia*
(4) *Sycon*

47) Match column-I with column-II and choose the correct option-

	Column-I		**Column-II**
(a)	Air bladder present	(i)	*Bangarus*
(b)	Mammary glands	(ii)	*Ornithorhynchus*
(c)	Neck absent	(iii)	*Bufo*
(d)	Skin cast	(iv)	*Betta*

(1) (a)-i, (b)-ii, (c)-iii, (d)-iv
(2) (a)-iv, (b)-ii, (c)-iii, (d)-i
(3) (a)-iv, (b)-iii, (c)-ii, (d)-i
(4) (a)-i, (b)-iii, (c)-ii, (d)-iv

48) Which of the following is incorrect statement regarding class-cyclostomata ?

(1) Cyclostomes have a sucking and circular mouth without jaws
(2) Cranium and vertebral column are bony
(3) Circulation is of closed type
(4) Their body is devoid of scales and paired fins

49) Stinging capsules (nematocysts) are found in :

(1) Wasp and physalia
(2) Adamsia and scorpion
(3) Sea pen and sea fan
(4) Meandrina and tapeworm

50) Read column-I, II & III and identify their correct match :-

	Column-I		**Column-II**		**Column-III**
(A)	Flatworms	(P)	Tissue level of organisation	(i)	Triploblastic Pseudocoelomate
(B)	Roundworms	(Q)	Organ level of organisation	(ii)	Diploblastic Radially symmetrical animals
(C)	Coelenterates	(R)	Cellular level of organisation	(iii)	Triploblastic, Acoelomate animals

(D)	Sponges	(S)	Organ system level of organisation	(iv)	Primitive multicellular, Mostly asymmetrical Animals

(1) A-Q-iii, B-S-i, C-P-ii, D-R-iv

(2) A-S-ii, B-Q-i, C-P-iii, D-R-iv

(3) A-Q-iii, B-S-i, C-P-iv, D-R-ii

(4) A-P-ii, B-Q-iii, C-S-i, D-R-iv

51) Cell junction that faciliates cell to cell communication is

(1) Tight junction

(2) Adhering junction

(3) Gap junction

(4) Interdigitation junction

52) Which of the followings are the main sound receptors of cockroach ?

(1) Anal style

(2) Anal cerci

(3) Elytra

(4) Fenestra

53) How many ova can lay by mature female frog at a time ?

(1) 2000 - 2500

(2) 2500 - 3000

(3) 2200 - 3200

(4) 1500 - 2000

54) Given below two statement, choose the correct answer from the options given below :-
Assertion : Liver secretes bile which is stored in gall gladder.
Reason : Bile emulsifies fat and pancreatic juices digest carbohydrate and protein.

(1) Both (A) and (R) are true and (R) is not the correct explanation

(2) (A) is true but (R) is false

(3) (A) is false but (R) is true

(4) Both (A) and (R) are true and (R) is the correct explanation of (A)

55) Which of the following statement is true for lymph?

(1) WBC + Serum + RBC

(2) Blood – (RBC + Platelets)

(3) RBC + Protein + Platelets

(4) WBC + RBC

56)

Which of the following animals has two separate circulatory pathways ?

(1) Frog
(2) Lizard
(3) Rat
(4) Labeo

57) **Statement-I :** Interstitial fluid has same mineral distribution as that in plasma.
Statement-II : Exchange of nutrients, gases etc. between the blood and the cells always occur through interstitial fluid.

(1) Both statement-I and II are correct.
(2) Both statement-I and II are incorrect.
(3) Statement-I is correct but II is incorrect.
(4) Only statement-I is incorrect.

58) Identify the incorrect statement.

(1) Leucocytes are nucleated.
(2) Neutrophils, eosinophils and basophils are different types of agranulocytes.
(3) Basophils are involved in inflammatory reactions.
(4) Eosinophils are associated with allergic reactions.

59) Read the statements carefully-
(i) If repeated checks of blood pressure of an individual is 140/90 or higher, it shows hypertension.
(ii) High blood pressure leads to heart diseases and also affects vital organs like brain and Kidney.
(iii) A special coronary system of blood vessels is present in our body exclusively for circulation of blood to and from the cardiac musculature.
(iv) Sequential events in the heart which is cyclically repeated is called the cardiac cycle.
Choose the correct statements according to given option -

(1) i & ii only
(2) iv only
(3) i only
(4) All i, ii, iii & iv

60) Which of the following factors are responsible for the left shift in the oxygen dissociation curve :-

(1) High pO_2, low pCO_2, low pH, high $[H^+]$
(2) High pCO_2, low pO_2, low temperature, high pH
(3) High pO_2, low temperature, high pH, low intermediate respiratory products
(4) High pCO_2, low pO_2, high temperature, low pH

61) Match the column-I and II.

	Column-I		Column-II
i.	Tidal volume (TV)	a.	1100-1200 ml
ii.	Inspiratory reserve volume (IRV)	b.	4800 ml
iii.	Residual volume (RV)	c.	500 ml
iv.	Vital capacity	d.	2500-3000 ml

Choose appropriate option :-

(1) i-a, ii-b, iii-c, iv-d

(2) i-c, ii-d, iii-a, iv-b

(3) i-c, ii-b, iii-d, iv-a

(4) i-a, ii-c, iii-d, iv-b

62) Which part of the brain stem is responsible for decrease the duration of Inspiration :

(1) Medulla oblongata

(2) Pons

(3) Cerebellum

(4) 1 and 3 both

63) **Assertion (A) :-** Nearly all the essential nutrients, and 70-80% of electrolytes and water are reabsorbed by PCT.
Reason (R) :- Tubular epithelial cells in different segment of nephron perform reabsorbed either by active or passive mechanism.

(1) A & R both are true and R is correct explanation of A.

(2) A is true but R is false.

(3) A is false but R is true.

(4) A & R both are true but R is not correct explanation of A.

64) Which blood vessel carries least amount of urea ?

(1) Pulmanory vein

(2) Renal artery

(3) Renal vein

(4) Hepatic vein

65) Read the column-I and II and match them correctly :-

	Column-I		Column-II
(A)	GH	(I)	Grave's disease
(B)	ADH	(II)	Diabetes mellitus
(C)	Thyroid hormone	(III)	Acromegaly

(D)	Insulin	(IV)	Diabetes insipidus
		(V)	Addison's disease

(1) A-III, B-IV, C-I, D-II

(2) A-III, B-II, C-I, D-IV

(3) A-V, B-IV, C-I, D-II

(4) A-III, B-IV, C-I, D-V

66) Pick out the incorrect statements ?

(1) Both myelinated and non-myelinated nerve fibres are enveloped with Schwann cells.

(2) Nissl's granules are absent in the axon and dendrites.

(3) Unmyelinated nerve fibres are commonly found in autonomic and somatic neural system.

(4) The axon transmits nerve impulses towards the synapse.

67) Association area not associated with :-

(1) Intersensory Association

(2) Memory

(3) Emotion & Sexual Behaviour

(4) Communication

68) Match the immune system component with its associated function :

Column-A		**Column-B**	
(A)	Thymosins	(1)	Provide cell-mediated immunity
(B)	T-lymphocytes	(2)	Serve as targets for immune responses
(C)	Antibodies	(3)	Promote the differentiation of T-lymphocytes
(D)	Antigens	(4)	Secretes peptide hormones crucial for immune system development
(E)	Thymus gland	(5)	Recognize and neutralize foreign substances

(1) (A)-(1), (B)-(4), (C)-(3), (D)-(2), (E)-(5)

(2) (A)-(2), (B)-(3), (C)-(1), (D)-(2), (E)-(5)

(3) (A)-(3), (B)-(1), (C)-(5), (D)-(2), (E)-(4)

(4) (A)-(4), (B)-(5), (C)-(2), (D)-(1), (E)-(3)

69) Which of the following is associated with increase in red blood cell count -

(1) ANF

(2) JGA

(3) GIP

(4) CCK

70) Regarding mechanism of FSH hormone, select the correct sequential steps :-
(A) Hormone receptor complex formation
(B) Biochemical-response
(C) Generation of cyclic-AMP
(D) Physiological-response
Choose the correct answer from the options given below :-

(1) A, D, C, B

(2) A, B, C, D

(3) A, C, B, D

(4) A, C, D, B

71) What is the primary reason for striated appearance of myofibril in muscle fibre ?

(1) Due to Sarcoplasmic reticulum storing calcium ion.

(2) The arrangement of multiple nuclei within the sarcoplasm.

(3) The distribution pattern of actin & myosin protein.

(4) The enclosing of muscle fibres by the sarcolemma.

72)

Fibrous joint	Cartilagenous joint	Synovial joint
↓	↓	↓
(A)	(B)	(C)

Choose the correct option :-

(1) A → Saddle joint, B → Pubic-symphysis, C → Knee joint

(2) A → Hinge joint, B → Knee joint, C → Pubic-symphysis

(3) A → Sutures, B → Inter vertebral disc, C → Knee joint

(4) A → Sutures, B → Knee joint, C → Hinge joint

73) Changes in GnRH pulse frequency in female is controlled by circulating level of :-

(1) Estrogen and Inhibin

(2) Progesterone and Relaxin

(3) Estrogen and progesterone

(4) Progesterone and inhibin

74) In test tube baby programme, the embryo more than 8 blastomeres is transferred to uterus, this is called as:

(1) ZIFT

(2) GIFT

(3) IUT

(4) ICSI

75) Which hormone in females is secreted only during the pregnancy?

(1) Estrogen

(2) Progesterone

(3) Human Chorionic Gonadotropin

(4) Oxytocin

76) **Statement-I :** FSH acts on the sertoli cells and stimulates secretion of some factors which help in the process of spermiogenesis.
Statement-II : Due to high concentration of estrogen during folliculogensis FSH increases more than the level of LH.

(1) Both **Statement I** and **Statement II** are correct.

(2) Both **Statement I** and **Statement II** are incorrect.

(3) **Statement I** is correct but **Statement II** is incorrect.

(4) **Statement I** is incorrect but **Statement II** is correct

77) Consider the following structures -
Womb, Clitoris, Fallopian tube, labia majora Mons pubis, uterine fundus
How many above structures are found in female as paired structure -

(1) Six

(2) Four

(3) Two

(4) Three

78) Given below are two statements are is labelled as Assertion and another is labelled as Reason
Assertion : In testis, the immature male germ cells (spermatogonia) produce sperms by spermatogenesis that begins at puberty.
Reason : A primary spermatocyte completes the first meiotic division (Reduction division) leading to formation of two equal haploid cell called secondary spermatoaftes, which have only 23 chromosome each.
In the light of the above statements select the correct answer from the options given below :

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(3) **Assertion** is correct but **Reason** is incorrect.

(4) **Assertion** is incorrect but **Reason** is correct.

79) The exaggerated response of the immune system to certain antigens present in the environment is called :-

(1) Active immunity

(2) Allergy

(3) Passive immunity

(4) Vaccine

80) Entamoeba histolytica is a protozoan parasite in the _____ of human which causes amoebiasis.

(1) Small intestine

(2) Liver

(3) Spleen

(4) Large intestine

81) Read the following statements and mark the incorrect option-

(1) Cocaine has a potent suppressive action on CNS.

(2) Cannabinoid receptors present principally in brain.

(3) Heroin generally taken by snorting and injection.

(4) Excessive dosage of cocaine causes hallucination.

82) Match list-I with list-II.

List-I		**List-II**	
(A)	Heroin	(i)	Interferes with the transport the neurotransmitter dopamine
(B)	Cannabinoids	(ii)	Very effective sedative and painkiller
(C)	Cocaine	(iii)	Known for their effects on cardiovascular system of the body.
(D)	Morphine	(iv)	A depressant and slows down body functions.

Choose the correct answer from the options given below-

(1) (A)-(iii), (B)-(iv), (C)-(ii), (D)-(i)

(2) (A)-(iv), (B)-(iii), (C)-(i), (D)-(ii)

(3) (A)-(ii), (B)-(i), (C)-(iii), (D)-(iv)

(4) (A)-(i), (B)-(ii), (C)-(iv), (D)-(iii)

83) Given below are two statements-

Statement-I : The symptoms of pneumonia include, fever, chills cough and headache.

Statement-II : Ringworm is one of the most common infections disease in men. Appearance of dry, scaly lesions on various parts of the body such as skin, Nails and scalps are the main symptoms of this disease.

In the light of the above statement choose the correct answer from the options given below-

(1) Statement-I is correct but II is incorrect.

(2) Statement-I is incorrect but II is correct.

(3) Statement-I and II both are correct.

(4) Statement-I and II both are incorrect.

84) During ice age, modern Homo sapiens arose between

(1) 75,000 - 10,000 years ago
(2) 100000 - 40000 years ago
(3) 18000 - 10000 years ago
(4) 75000 - 40000 years ago

85) In recombinant DNA methods, the term vector refers to :-

(1) The enzyme that cuts DNA into restriction fragments
(2) The sticky end of a DNA fragment
(3) A plasmid used to transfer DNA into a living cell
(4) A DNA probe used to identify a particular gene

86) Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R) :
Assertion (A) : Restriction exonuclease are also called 'molecular scissors'.
Reason (R) : When fragments generated by restriction endonuclease are mixed, they join together due to their sticky ends.
In the light of the above statements, choose the correct answer from the options given below:

(1) Both A and R are true and R is the Correct Explanation of A.
(2) Both A and R are true and R is NOT the Correct Explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

87) Main processes of recombinant DNA technology are given below. Arrange these processes in a correct sequence-
(A) Obtaining foreign gene product
(B) Isolation of genetic material
(C) Cutting of genetic material by restriction endonuclease
(D) Amplification of desired gene using PCR
(E) Downstream processing
(F) Insertion of recombinant DNA into host organism

(1) ABCDEF
(2) CABDEF
(3) EFBDCA
(4) BCDFAE

88) Which of the following peptides are present in mature insulin?

(1) A-peptide and B-peptide
(2) A-peptide and C-peptide
(3) B-peptide and C-peptide

(4) Only C-Peptide

89)

Match List I with List II :

List I		**List II**	
A.	α-1 antitrypsin	I.	Cotton bollworm
B.	Cry IAb	II.	ADA deficiency
C.	Cry IAc	III.	Emphysema
D.	Enzyme replacement therapy	IV.	Corn borer

Choose the correct answer from the options given below:

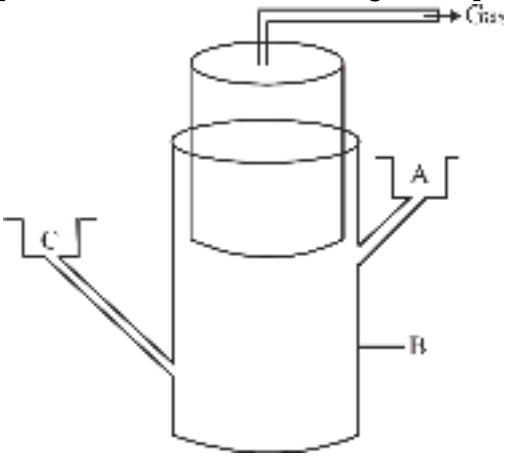
(1) A-II, B-I, C-IV, D-III

(2) A-III, B-I, C-II, D-IV

(3) A-III, B-IV, C-I, D-II

(4) A-II, B-IV, C-I, D-III

90) The below diagram shows a typical bio gas plant. Which of the following four option, products



labelled as A, B and C are correctly identified ?

(1) A-Sludge, B–Digester, C–Dung, water

(2) A–Digester, B–Methane, Carbon dioxide, C–Dung, water

(3) A–Sludge, B–Ethylin, Carbon dioxide, C–Dung, water

(4) A–Sludge, B–Methane, Carbon dioxide, C–Sewage

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	3	3	4	4	3	3	3	4	1	3	2	3	4	2	4	2	1	2	2	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	4	4	2	3	1	4	2	3	3	2	1	1	2	3	3	4	1	4	4
Q.	41	42	43	44	45															
A.	1	3	3	1	4															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	1	4	3	1	2	4	1	3	2	2	2	3	2	3	4	1	1	2	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	3	4	3	4	2	3	3	2	3	3	1	2	4	3	2	4	1	3	4
Q.	86	87	88	89	90															
A.	1	3	3	2	2															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	4	1	4	2	2	1	1	4	2	3	2	2	4	4	1	3	2	4	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	2	4	3	1	2	3	2	3	3	2	1	3	4	1	3	2	2	1	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	1	2	3	2	1	2	2	3	1	3	2	2	1	2	3	1	2	4	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	2	2	4	3	1	2	3	3	2	3	3	3	3	3	3	3	3	2	2	4
Q.	171	172	173	174	175	176	177	178	179	180										
A.	1	2	3	1	3	4	4	1	3	1										

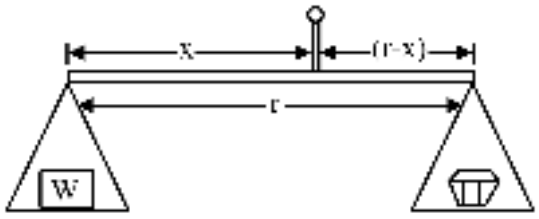
SOLUTIONS

PHYSICS

1) Least count (LC) = $\left(\frac{b-a}{b}\right)$ MSD ;
here a = 9,
b = 10 ; 1 MSD = 1 mm

LC = $\left(\frac{10-9}{10}\right)$ 1 mm = 0.1 mm

There is zero error in measurement.
As the zero of the vernier lies to the left of the main scale, it has negative zero error. For negative zero error, the coinciding division is to be read from right and it has 5th division coinciding from right.
Thus zero error = ZE = –5 × 0.1 mm = –0.5 mm = –0.05 cm
Reading at measurement : MSR = 8.5 cm
VSR = 8th VSD × 0.1 mm = 0.08 cm
Corrected reading = MSR+VSR × LC – zero error
= 8.5 cm + 0.08 cm – (–0.5mm) = 8.63 cm

2) 

Let true weight of object is W
(1) When object placed in left pan and 9 kg weight in right pan
W × x = 9g (r–x) .....(1)
(1) When object place in right pan and 9 kg weight in left pan
16g(x) = W(r–x) .....(2)
Equation (1) / (2)

$\Rightarrow \frac{W}{16} = \frac{9}{W} \Rightarrow W^2 = 9 \times 16$

$\Rightarrow W = \sqrt{144} \Rightarrow W = 12$ kg

3) From graph
Y = slope of stress-strain graph = 1×10^{11} N/m^2
At strain 4×10^{-4}

Energy density = $\frac{1}{2}$ × Stress × Strain = $\frac{1}{2}$ × Y × (Strain)^2

= $\frac{1}{2}$ (10^{11}) $(4 \times 10^{-4})^2$ = $\frac{1}{2}$ (10^{11}) (16×10^{-8}) = 8 kJ/m^3

4) $d_1 = 2r_1$ = 2 mm ; $d_2 = 2r_2$ = 5 mm
If h_1 and h_2 are the rise of water in two limbs ; then due to effect of surface tension, difference

in height

$\Delta h = h_1 - h_2$ also $\left[h \propto \frac{1}{r}\right]$, $d_2 = 2.5$ m

$= \frac{2T\cos\theta}{\rho g}\left[\frac{1}{r_1} - \frac{1}{r_2}\right] = \frac{(2)(7.2 \times 10^{-2})}{(10^3)(10)}\left[\frac{1}{2} - \frac{1}{5}\right](10^3)(2)$

= 8.64 mm

5) In meter bridge experiment, a high resistance is connected in series to protect it, from damage due to large currents especially when the bridge is largely unbalanced. (before null point) ⇒ (A) correct. According to Ohm's law, if resistance of circuit increases, then current through circuit decreases for same supply ⇒ (R) wrong

6)

For the peak voltage of input waveform, both zener diodes will go under breakdown. For positive cycle zener 1 is FB & zener 2 will be reverse biased, for V > |10V| zener 2 will undergo breakdown so output for positive cycle will be clipped remains constant at 10 V. For negative cycle zener 1 will be reverse biased and zener 2 is FB, for V > |6V| zener 1 will undergo breakdown so output for negative cycle will be clipped at 6V.

7) Response (i) and (ii) suggest presence of a unidirectional device like diode.
Response (iii) and (iv) suggest non polar linear behaviour of current, possible in resistor.
Response (v) and (vi) suggest charging of capacitor.

8) $F \propto x^{-1/3}$

$\Rightarrow a \propto x^{-1/3}$

$\Rightarrow \int v dv \propto \int x^{-1/3} dx$

$\Rightarrow v^2 \propto x^{2/3}$

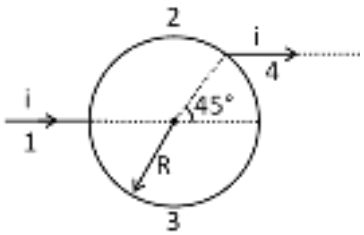
$\Rightarrow v \propto x^{1/3}$

power = F.v

$p \propto x^{-1/3}.x1/3$

$\Rightarrow$ power = const. $\propto x^0$

9)

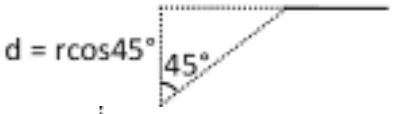


Calculation:

Magnetic field at centre = $B_1 + B_2 + B_3 + B_4$

$B_1 = 0$ (wire passes from point)

$B_2 + B_3 = 0$ (both field cancel each other)



$\frac{\mu_o i}{4\pi r \cos 45^\circ}(\sin 90^\circ - \sin 45^\circ)$

$B_4 = \frac{\frac{\mu_o i}{4\pi R\frac{1}{\sqrt{2}}}\left(1-\frac{1}{\sqrt{2}}\right)}{}$

$B_4 = \frac{\mu_o i}{4\pi R}\sqrt{2}\left(1-\frac{1}{\sqrt{2}}\right)$

$B_4 = \frac{\mu_o i}{4\pi R}\left(\sqrt{2}-\frac{\sqrt{2}}{\sqrt{2}}\right)$

$B_4 = \frac{\mu_o i}{4\pi R}\left(\sqrt{2}-1\right)$

10) $x_m \propto \frac{1}{T}$

$\frac{(\chi m)_{300K}}{(\chi m)_{350K}} = \frac{350}{300}$

or $(\chi_m)_{300K} = \frac{350}{300} \times 2.8 \times 10^{-4} = 3.267 \times 10^{-4}$

11)

By energy conservation, $E_1 + E_2 = E_{mix}$

$\Rightarrow \frac{n_1 f_1 RT_1}{2} + \frac{n_2 f_2 RT_2}{2} = \frac{(n_1 + n_2) f RT_{mix}}{2}$

$\Rightarrow \frac{3}{2}P_1V_1 + \frac{3}{2}P_2V_2 = \frac{3}{2}P_{mix}(V_1 + V_2)$

$\Rightarrow \frac{3}{2} \times 1 \times 2 + \frac{3}{2} \times 2 \times 3 = \frac{3}{2} \times P_{mix}(5)$

$\therefore P_{mix} = \frac{8}{5} = 1.6$ atm.

12) For series combination,

$k_s = \frac{2k_1k_2}{k_1 + k_2} = \frac{2.k.2k}{k + 2k}$

$k_s = \frac{4k}{3}$

For parallel combination,

$k_p = \frac{k_1 + k_2}{2} = \frac{K + 2k}{2}$

$k_p = \frac{3k}{2}$

$\frac{k_s}{k_p} = \frac{4k}{3} \times \frac{2}{3k} = \frac{8}{9}$

13) $(Q_{extracted}) = Q_{BC} + Q_{CD}$

$= nC_V\Delta T + nC_P\Delta T$

$= n\frac{fR}{2}\Delta T + n\left(\frac{f}{2}+1\right)R\Delta T$

$= \frac{3}{2}V^0\Delta P + \frac{5}{2}P'\Delta V$

$= \frac{3}{2}(V_0 2P_0) + \frac{5}{2}(3P_0)2V_0$

$= 18\ P_0V_0$

Efficiency (η%)

$\eta\% = \frac{W.D.}{Q_+} \times 100$

$= \frac{4P_0V_0}{18P_0V_0} \times 100$

$= 22.22\ \%$

14) $\frac{dT}{dt} = \frac{-e\sigma A}{mS}\left(T^4 - T_s^4\right)$

$\frac{dT}{dt} \propto \frac{A}{m} \propto \frac{\pi r^2}{\rho \times \left(\frac{4}{3}\pi r^3\right)}$

$\frac{dT}{dt} \propto \frac{1}{\rho r}$

$\frac{(dT/dt)_1}{(dT/dt)_2} = \frac{\rho_2 r_2}{\rho_1 r_1} = \frac{1}{2} \times \frac{2}{1} = \frac{1}{1}$

15) $Z = \sqrt{(R)^2 + (X_L - X_C)^2}$

$R = 10\Omega$, $X_L = \omega L = 2000 \times 5 \times 10^{-3} = 10\Omega$

$X_C = \frac{1}{\omega C} = \frac{1}{2000 \times 50 \times 10^{-6}} = 10\Omega$ i.e. $Z = 10\Omega$

Maximum current $i^0 = \frac{V_0}{Z} = \frac{20}{10} = 2A$

Hence $i_{rms} = \frac{2}{\sqrt{2}} = 1.4A$

and $V_{rms} = 4 \times 1.41 = 5.64$ V

16) Energy of the skylab in the first orbit is $-\frac{GMm}{2(2R)} = -\frac{GMm}{4R}$

Total energy required to place the skylab into the orbit of radius 2R from the surface of earth is

$-\frac{GMm}{4R} - \left(-\frac{GMm}{R}\right) = \frac{3GMm}{4R}$

$= \frac{3gR^2m}{4R} = \frac{3}{4}mgR$ Energy of the skylab in the second orbit = –(GMm)/6R. Energy needed to shift the skylab from the first orbit to the second orbit is

$-\frac{GMm}{4R} - \frac{GMm}{6R} = \frac{GMm}{R} \times \frac{2}{24} = \frac{mgR}{12}$

17) $U_{max} = U^0 + \frac{1}{2}KA^2$

$\frac{1}{2}KA^2 = U_{max} - U_0$

$= 20 - 4 = 16$

(a) $U = U^0 + \frac{1}{2}Kx^2$

$U = 4 + \frac{1}{2}K\frac{A^2}{4}$

$U = \frac{4 + \frac{16}{4}}{}$

$U = 8\ J$

(b) $KE = \frac{1}{2}K\left(A^2 - \frac{A^2}{16}\right) = \frac{15}{16}\left(\frac{1}{2}KA^2\right)$

$= \frac{15}{16} \times 16 = 15J$

(c) $KE = \frac{1}{2}KA^2 = 16J$

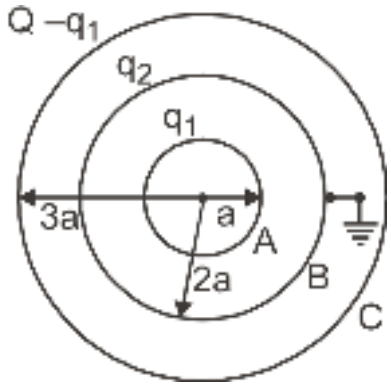
(d) $KE = \frac{1}{2}K\left(A^2 - \frac{A^2}{4}\right) = \frac{3}{4}\left(\frac{1}{2}KA^2\right)$

$= \frac{3}{4} \times 16$

$= 12\ J$

So, a → R ; b → S ; c → Q ; d → P

18)



From given conditions,

$V_A = V_C$ and $V_B = 0$

$\Rightarrow V_B = \frac{K(Q - q_1)}{3a} + \frac{Kq_2}{2a} + \frac{Kq_1}{2a} = 0$

$\Rightarrow 2Q + q_1 + 3q_2 = 0$ ....(1)

Using $V_C = V_A$

$\frac{K(Q - q_1)}{3a} + \frac{Kq_2}{3a} + \frac{Kq_1}{3a}$

$= \frac{Kq_1}{a} + \frac{K(Q - q_1)}{3a} + \frac{Kq_2}{2a}$

$\Rightarrow q_1 = -\frac{q_2}{4}$ ....(2)

Using it in (1),

$$\Rightarrow q_2 = -\frac{8Q}{11}$$

19) **1. Flux through the cube:**
Each charge at the vertex of the cube is shared by 8 neighboring cubes, meaning the total charge enclosed inside the given cube is:

$$Q_{enclosed} = \frac{8q}{8} = q$$

Using Gauss's law: $\Phi_{cube} = \frac{q}{\epsilon_0}$
So, **Statement 2 is correct.**
2. Flux through a single face:
The total flux is uniformly distributed across six faces of the cube:

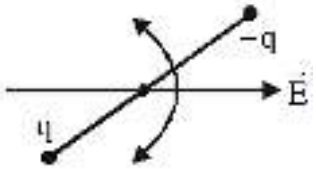
$$\Phi_{face} = \frac{\Phi_{cube}}{6} = \frac{q}{6\epsilon_0}$$

This matches Statement 3, so **Statement 3 is correct.**
3. Flux through surface ABCD:
The given flux is $\frac{q}{24\epsilon_0}$, which is incorrect because it should be $\frac{q}{6\epsilon_0}$. So, **Statement 1 is incorrect.**

20) Time taken to become parallel to the field

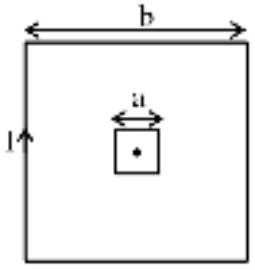
$$t = \frac{\text{Time period}}{4}$$

$$= \frac{T}{4} = \frac{1}{4}\left[2\pi\sqrt{\frac{I}{PE}}\right]$$



$$= \frac{\pi}{2}\left[\sqrt{\frac{ML^2/2}{qLE}}\right] = \frac{\pi}{2}\left[\sqrt{\frac{ML}{2qE}}\right]$$

21)

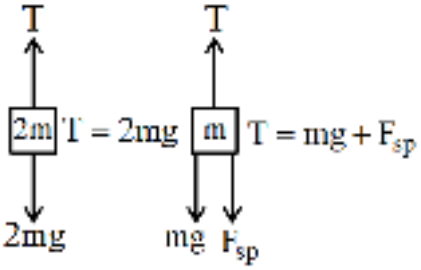


$$B = \left[\frac{\mu_0}{4\pi}\frac{I}{b/2} \times 2\sin 45\right] \times 4$$

$$\phi = 2\sqrt{2}\frac{\mu_0}{\pi}\frac{I}{b} \times a^2$$

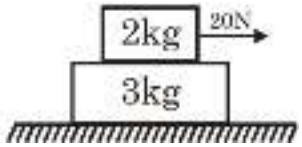
$$\therefore M = \frac{\phi}{I} = \frac{2\sqrt{2}\mu_0 a^2}{\pi b} = \frac{\mu_0}{4\pi} 8\sqrt{2}\frac{a^2}{b}$$

22) Just after the string is cut, string force (tension) vanishes immediately, but the spring cannot regain its natural length instantaneously. In consequence the spring force remains the same just before and after the string ruptures. Hence, the bodies connected to the spring experience same spring force just after cutting the string. FBD's before cutting the string :



$2mg = mg + F_{sp} \Rightarrow F_{sp} = mg$
After the string is cut, T vanishes. So immediately after string is cut,
$\Rightarrow a_m = 2g \downarrow$ and $a_{2m} = g \downarrow$.

23)



If both move together



$a_{common} = 4$ m/sec^2
Maximum friction force between 2kg & 3kg block = μ mg = 0.8 × 20 = 16 N
Force required to move 3 kg block = 3 × 4 = 12 N
So $F_{requires} \leq$ friction force, So both block will move together with acceleration 4 m/s^2 and friction force between blocks is 16 N.

24)

$(\mu_1 - 1)t_1 = (\mu_2 - 1)t_2$
$(1.25 - 1)\, t_1 = (1.5 - 1)t_2$

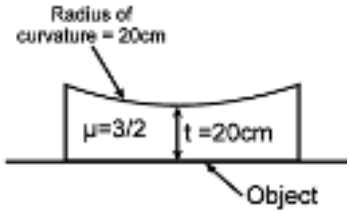
$$\frac{t_1}{t_2} = \frac{0.5}{0.25} = \frac{2}{1}$$

25) Given : $f = KR^a \rho^b G^c$ ...(i)
Where,
$[T^{-1}] = f$ = vibration frequency
$[L] = R$ is the radius of star,
$[ML^{-3}] = \rho$ is the density of star,

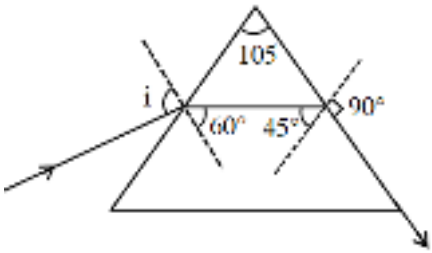
$$[M^{-1}L^3T^{-2}] = \frac{Fr^2}{m_1 m_2} = G$$

is the universal gravitational constant,

k is a constant.
by putting all the dimensions in equation (i) we get
$[M^0 L^0 T^{-1}] = [L]^a [ML^{-3}]^b [M^{-1} L^3 T^{-2}]^c$
By comparing both sides equation we get,
b –c = 0
a – 3b + 3c = 0
–2c = – 1
$c = \frac{1}{2}$; $b = \frac{1}{2}$; a = 0
Then $b + c = \frac{1}{2} + \frac{1}{2} = 1$



26) Considering refraction at the curved surface,
u = –20 ; μ_2 = 1
μ_1 = 3/2 ; R = +20
applying $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$
$\Rightarrow \frac{1}{v} - \frac{3/2}{-20} = \frac{1-3/2}{20} \Rightarrow v = -10$
i.e. 10 cm below the curved surface of 10 cm above the actual position of flower.



27) $\sin i_c = \frac{1}{\mu} = \frac{1}{\sqrt{2}}$
$i_c = 45°$
$A = r_1 + r_2$
$105 = r_1 + 45$
$r_1 = 60$
$\sqrt{2} = \frac{\sin i}{\sin 60}$
$i = \sin^{-1}\left(\sqrt{\frac{3}{2}}\right)$

28)

$L = v_o + f_e$
$v_O = L - f_e = 19 - 3 = 16$ cm
$M = [(f_o - v_O) / f_0)] \times (25 / f_e)$
$M = [(2 - 16) / 2] \times (25 / 3)$
$M = 175/3$

29) $\vec{C} = \left(\frac{\vec{A}.\vec{B}}{B}\right)\hat{A}$

$= \left(\frac{2-3-4}{3}\right)\left(\frac{i+3j-2k}{\sqrt{14}}\right)$

$= \frac{-5(i+3j-2k)}{\sqrt{14}}$

30) Since $\frac{f}{2} < f_0$ i.e. the incident frequency is less than threshold frequency. Hence there will be no emission of photoelectrons.

$\Rightarrow$ current $= 0$

31) $I = \frac{e}{T} = \frac{e\omega}{2\pi} = \frac{eV}{2\pi r}$

$I = \frac{1.6\times10^{-19}\times2.2\times10^{5}\times7}{2\times22\times0.5\times10^{-11}}$

$= 1.12$ mA

$= 112 \times 10^{-2}$ mA

32) $\tilde{v} = 6\hat{i} + (20-4t)\hat{j}$

time when $v_y = 0$

$20 - 4t = 0$

$t = 5$

$T = 2 \times t = 10$ sec

$\tan 53 = \frac{20-4t}{6}$

$\frac{4}{3} = \frac{20-4t}{6}$

$t = 3$ sec

$R = v_x \times T$

$= 6 \times 10 = 60$ m

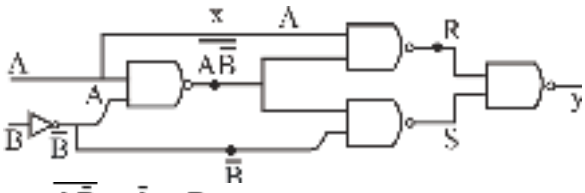
$H = \frac{v_y^2}{2\times a} = \frac{20\times20}{2\times4} = \frac{400}{8} = \frac{200}{4} = 50m$

33) $-H = 4t_1 - \frac{1}{2}gt_1^2$

$-H = -4t_2 - \frac{1}{2}gt_2^2$

$-H = -\frac{1}{2}gt^2$ by solve $t = \sqrt{t_1 t_2}$

34)



$x = \overline{A\bar{B}} = \bar{A} + B$

$R = \overline{x \cdot A} = \overline{(\bar{A} + B) \cdot A} = \overline{A\bar{A} + AB} = \overline{AB}$

$S = \overline{B} \cdot x = \bar{B} \cdot (\bar{A} + B)$

$\overline{\bar{B} \cdot \bar{A} + \bar{B} \cdot B} = A + B$;

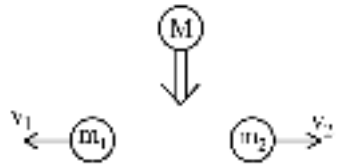
$y = \overline{R.S} = \overline{A\bar{B}} \cdot (A + B)$

$= AB + \left(\overline{A + B}\right)$

$= AB + \bar{A}\,\bar{B}$

35) By COLM

$|P_1| = |P_2|$



$\lambda = \frac{h}{p}$

$\frac{\lambda_1}{\lambda_2} = \frac{h/p_1}{h/p_2} = 1 : 1$

36) For pressure we consider only vertical distances means height, so pressure at piston = P

$P = \frac{F}{A} = (\rho_s h_2 + \rho_3 h_3)g$

37) $\vec{E} \cdot \vec{B} = 0$ and direction of propagation (–Z) = $\vec{E} \times \vec{B}$

38)

$R_1 = \frac{(100)^2}{25} = 400\Omega$; $R_2 = \frac{(50)^2}{50} = 50\Omega$

$R_3 = \frac{(200)^2}{100} = 400\Omega$; $R_4 = \frac{(50)^2}{25} = 100\Omega$

$V_1 = \left(\frac{400}{400 + 50}\right) V = \frac{8}{9}V$;

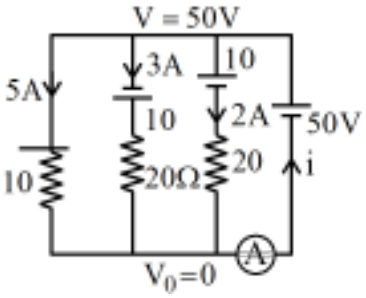
$V_2 = \left(\frac{50}{400 + 50}\right) V = \frac{V}{9}$

$V_4 = \left(\frac{100}{400 + 100}\right) V = \frac{V}{5}$,

$V_3 = \left(\frac{400}{400 + 100}\right) V = \frac{4V}{5}$

brightness of bulb ∝ power loss ∝ v^2/R

39)



i = 5 + 3 + 2 = 10 A

40)

$J = \frac{I}{A}$ and $E = \rho J \Rightarrow E = \rho\frac{I}{A}$

(I = current, ρ = resistivity, A = Area of cross section)

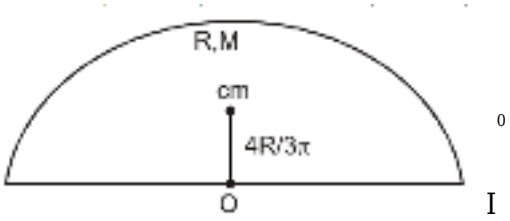
$\Rightarrow E \propto \frac{1}{A}$ (for constant I and ρ)

• Assertion (A) – true : $E \propto \frac{1}{A}$ for constant current.

• Reason (R) – True : $\vec{E} = \rho\vec{J}$ is correct.
(Microscopic view of ohm's law)

• Reason correctly explains the assertion

41)



$I = I_{CM} + Md^2$

$\frac{1}{2}MR^2 = i_{CM} + M\left(\frac{4R}{3\pi}\right)^2$

$\Rightarrow I_{CM} = MR^2\left(\frac{1}{2} - \frac{16}{9\pi^2}\right)$

42)

using angular momentum conservation

$L_i = 0$

$L_t = mvR - I\omega$

$mvR = I.\omega$

$\omega = \left(\frac{1}{2}\right)$

$(v + \omega R)t = 2\pi R$

$\left(1 + \frac{1}{2} \times 2\right) = 2\pi \times 2$

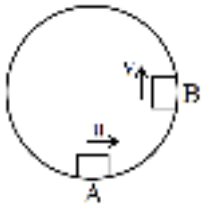
t = 2π sec.

43) $V = \sqrt{\frac{T}{\mu}}$

$V = f\lambda$

$\frac{\lambda_2}{\lambda_1} = \frac{V_2}{V_1} = \sqrt{\frac{T_2}{T_1}}$

$\lambda_2 = 0.08\sqrt{\frac{20g}{5g}} = 0.16m$

44) 

$u = \sqrt{5gR}$

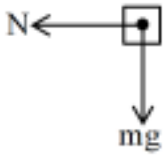
$v = \sqrt{3gR}$

At B :-

$a_c = \frac{V^2}{R}$

$\Rightarrow a_c = 3g$

FBD of block at B :-



$F_t = ma_t$

$\Rightarrow mg = ma_t$

$\Rightarrow a_t = g$

$a_{net} = \sqrt{a_c^2 + a_t^2}$

45) At steady state capacitor → open P →Q (KVL)

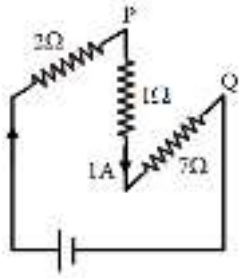
$I = \frac{10}{10} = 1A$

$V_P - 1 - 7 = V_Q$

$V_P - V_Q = 8V$

$U = \frac{1}{2}CV^2 = \frac{1}{2} \times 4 \times 10^{-6} \times (8)^2$

$= 128\ \mu J$



CHEMISTRY

46) $\frac{r_1}{r_2} = \left(\frac{n_1}{n_2}\right)^2 = \frac{r}{4r}$

$\frac{n_1}{n_2} = \frac{1}{2}$

$$\frac{T_1}{T_2} = \left(\frac{n_1}{n_2}\right)^3 \left(\frac{z_2}{z_1}\right)^2 = \left(\frac{1}{2}\right)^3 \left(\frac{1}{1}\right)^2 = \frac{1}{8}$$

47) $(I.E.)_H = -(E_1)_H = x = 13.6 \times (1)^2$

$E \propto Z^2 \quad \Delta E = E_6 - E_3 = 13.6 \times (3)^2 \left[\frac{1}{3^2} - \frac{1}{6^2}\right]$

48) At t = 3 min, according to graph
$[H_2]$ increases first ie, H_2 was added to the equilibrium

49)

$NH_4OH + NH_4Cl$ is basic buffer solution

So, $pOH = pK_b + \log \frac{[Salt]}{[Base]}$

or $[OH^-] = K_b \times \frac{[NH_4OH]}{[NH_4^+]}$

50)

(a) 1 mole water = 18 g = 18 mL → (iii)
(b) 1 g molecule O_2 at NTP = 22.4 L → (i)
(c) 12.022×10^{23} molecules N_2H_4 = 2 moles
⇒ 8 H atoms per molecule → total 8 g atoms H → (iv)
(d) 0.16 g CH_4 → 0.16/16 = 0.01 mole
⇒ neutrons in CH_4 (6 neutrons per molecule) = 0.01 × 6 = 0.06 mole neutrons → (ii)

51)

$$A^{2+} + KMnO_4 \rightarrow \overset{+5}{\underline{A}}O_3^-$$

$n_f \quad 3 \quad n_f = ?$

2.7×10^{-3} mol $\quad 1.6\times10^{-3}$ mol

no. of equivalents of A^{2+} = no. of equivalents of $KMnO_4$
$2.7 \times 10^{-3} \times 3 = 1.6 \times 10^{-3} \times n_f$
$n_f = 5$ ⇒ $KMnO_4$ goes in to Mn^{2+}
⇒ medium is acidic

52) $\Delta H = \Sigma(\Delta BE)_{reactants} - \Sigma(\Delta BE)_{products}$
$\Delta H = [4 \times e_{C-H} + e_{H-H} + e_{C=C}] - [6 \times e_{C-H} + e_{C-C}]$
$\Delta H = [4 \times 414 + 435 + 615] - [6 \times 414 + 347]$
$\Delta H = -125$ kJ

53) $HCl + NaOH \rightarrow NaCl + H_2O \quad \Delta H = -x$
Milli eq. of HCl = 10 × 1 × 1 = 10
$H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O \quad \Delta H = -y$
Milli eq. of H_2SO_4 = 10 × 2 × 1 = 20

For double equivalents of H_2SO_4, double heat will release. So. y = 2x

54) $W = -P_{ext} \cdot \Delta V$
$W = -(2 \times 10^5)(500 \times 10^{-6}) = -100$ J
$q = +200$ J
$\Delta E = q + w$
$\Delta E = 200 + (-100)$
$= 100$ J

55)

$$\text{Solubility} \propto \frac{1}{K_H}$$

$K_H \downarrow \rightarrow$ Solubility $\uparrow$
B > C > A

56)

• Chloroform + Acetone show negative deviation due to strong hydrogen bonding (A–B interactions > A–A or B–B).
• For an ideal solution, $\Delta H_{mix} = 0$, $\Delta V_{mix} = 0$, but $\Delta G_{mix} < 0$ and $\Delta S_{mix} > 0$.

57) $In^+ \rightarrow In^{+3} + 2e^-$; 0.42
$In^{+2} + e^- \rightarrow In^+$; –0.40
_____
$In^{+2} \rightarrow In^{+3} + e^-$; 0.44

$$E^o_{cell} = E^o_{Cu^{+2}/Cu^+} - E^o_{In^{+3}/In^{+2}}$$
$\Rightarrow 0.15 - (-0.44) = 0.59$

$$E = E^o - \frac{0.0591}{n} \log Q$$

At eq.

$$E^o = \frac{0.0591}{n} \log k_{eq}$$

$$0.59 = \frac{0.0591}{1} = \log k_{eq}$$

$k_{eq} = 10^{10}$

58)

$2H^+ + 2e^- \rightarrow H_2(g)$

$$E_{RP} = E^0_c - \frac{0.0591}{2} \log \frac{1}{[H^+]^2}$$

$$E_{RP} = 0 + \frac{0.0591}{2} \log [H^+]^2$$

$E_{RP} \Rightarrow -0.059$ pH

$(pH)_1 = 0 \quad E_1 = 0$ V
$(pH)_2 = 7 \quad E_2 = -0.059 \times 7 = -0.413$ V

Reduction potential decrease by 0.413 V i.e. O.P $\uparrow$ es

59)

Given mechanism :
(1) $A \xrightarrow{k_1} B + X$ (slow)
(2) $A + X \xrightarrow{k_2} B + C$ (fast)
Since step (1) is slow, it is rate-determining.
Hence, rate law, $r = k_1[A]$ → first order in A.
Therefore, statement 'Order of the reaction with respect to A is two' is incorrect.

60)

$r = k[A]^2[B]^{-1}$
If [A] & [B] are doubled
$r' = k[2A]^2[2B]^{-1}$
$r' = 2r$

61) Concept

62)

NCERT-XI, Pg. No. # 93, OLD NCERT, Part-2, p-block

63)

$I_2Cl_6 \rightarrow \vec{\mu} = 0$ (Non polar)

64)

NCERT XI # 118, 119

65) NCERT-XI Pg. # 112, Part-I, Chapter-3

66)

Structure: H–O–N →O

A. (a) H: Valence = 1, bonds = 1 → F.C. = 1 − (0 + ½×2) = 0
B. (b) N: Valence = 5, bonds = 4 (2 with O–N, 2 with N=O) → F.C. = 5 − (0 + ½×8) = +1
C. (c) O: Valence = 6, bonds = 1 → F.C. = 6 − (6 + ½×2) = −1

Final Answer: Option (4)

67)

A complex is Colourless when no d–d transition is possible. In $K_3[Cu(CN)_4]$, copper is in +1 oxidation state with d^{10} configuration, so d–d transition do not occur. Other given complexes contain partially filled d–orbitals and hence are coloured.
Final Answer: Option (3)

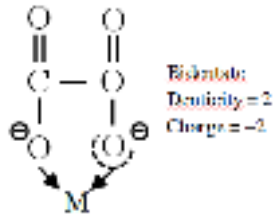
68)

Solution:

A. Ferrocene: Fe sandwiched between two cyclopentadienyl rings → contains meatal-carbon bond → organometallic.
B. Ziese's $K[PtCl_3(H^2–C_2H_4)]$ salt : bonded to carbon → organometallic.
C. Grignard reagent (RMgX): → Carbon-magnesium bond → organometallic.
D. Cis-platin $[Pt(NH_3)_2Cl_2]$: Platinum bonded to NH_3 and Cl^-→ no direct metal-carbon bond → not organometallic.

Final Answer: Option (4)

69) Is correct answer
Oxalato $(C_2O_4^{2-})$



Ethylene diamine triacetato $(EATA)^{-3}$ = Pentadentate
Charge = –3
Dimethyl glyoximato $(dmg)^{-1}$ = Pentadentate
Charge = – 1
triphenyl phosphire $(P(Ph)_3]$ = denticity = 1
Charge = 0
Diethylene triamine (dien) = denticity = 3
Charge = 0

70) **Solution**
All group-16 show allotropy → true.
O_2 is diatomic gas, S_8 is solid → difference in mp & bp explained by atomicity → true.
–2 oxidation state stability decreases from O → Po true.
Hence all are correct.
Final Answer - (Option 4)

71)

Due to inert pair effect.
Pb^{+4} strong O.A.

I^- strong R.A.

72)

X = F, Y = Cl
↓
Strong oxidising agent oxidises 'H_2O' into O_2
Cl_2 → Undergoes disproportionation.

73)

ZnS - White

74) like f^2 and $f^{(14-2)}$ having same number of unpaired e^- is equal = 2
means same magnetic nature.

75) $CaCrO_4$ & $SrCrO_4$
more soluble
 $BaCrO_4$ gives ppt. first

76)

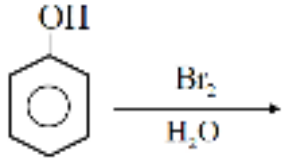
A. Initial H_2SO_4 : 100 mL × 1 N = 100 meq.
B. Titration of residual acid: 20 mL diluted solution needs 32 mL of 0.1 N NaOH . *Meq* in 20 mL = 32 × 0.1 = 3.2 meq.
C. Total residual acid in $500\ mL : \frac{3.2}{20} \times 500 = 80 meq.$
D. Acid used by NH_3 : 100 – 80 = 20 meq.

$$\%N = \frac{1.4 \times \text{ Meq of acid used}}{\text{Mass of compound}} = \frac{1.4 \times 20}{1.216} \approx 23\%$$

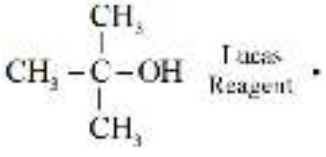
Final Answer: Option (3) 23%

77) $CH_3 - CH_2 - OH \xrightarrow[NaOH]{I_2}$ Yellow ppt

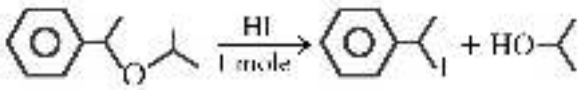
$CH_3 - CH_2 - NH_2 \xrightarrow[KOH]{CHCl_3}$ Foul smell of isocyanide



White ppt

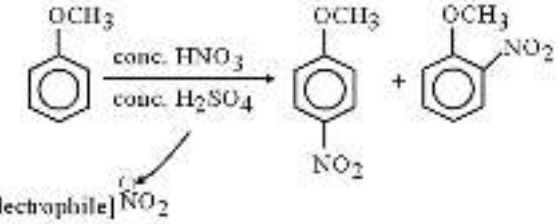


3° Alcohol react fastest

78) 

79)

⇒ +I / + M groups are ortho and para directing groups.



⇒ [Electrophile]

 Option 4 is correct

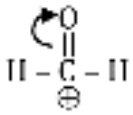
Final answer : 4

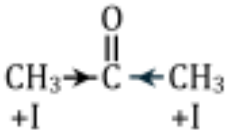
80)

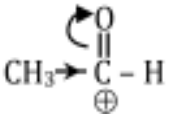
Leaving group tendency ∝ stability of anion.

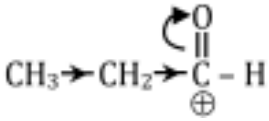
81)

Reactivity towards NAR ∝ Positive charge on carbonyl carbon $\propto \frac{\varepsilon WG}{\varepsilon DG} \propto \frac{1}{\text{Steric hindrance}}$

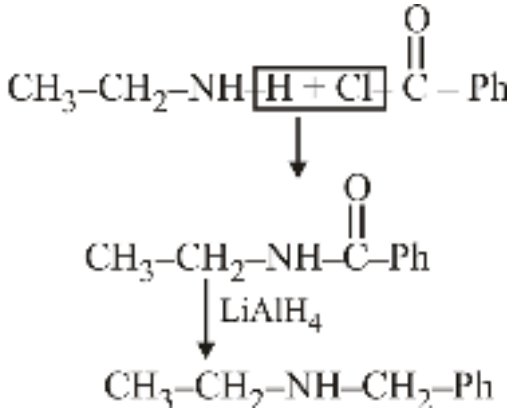
(P) 
More positive on carbonyl carbon (Most reactive)

(Q) 
Positive ↓es on carbonyl carbon due to +I (Least reactive)

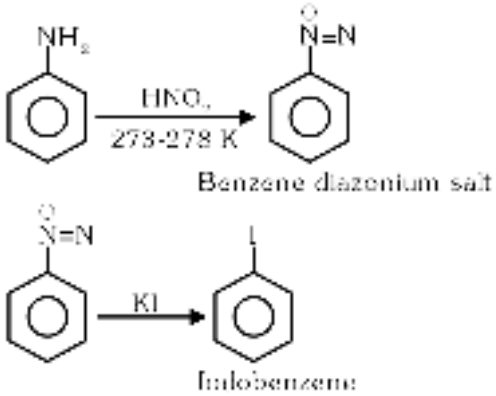
(R) 
Positive ↓ es carbonyl carbon

(S) 
Positive ↓es on carbonyl carbon

Final Answer: Option (2)

82) 

83) Benzenediazonium chloride is a colourless crystalline solid. It is readily soluble in water and is stable in cold but reacts with water when warmed. It decomposes easily in the dry state.



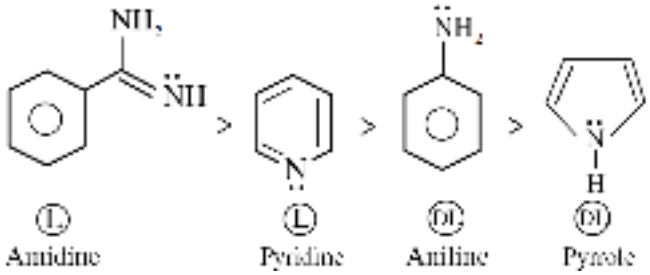
84)

Sucrose does not show mutarotation.

85)

Due to resonance C–N bond gets double bond character
Reso↑, double bond character↑, BL↓

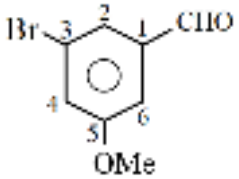
86)



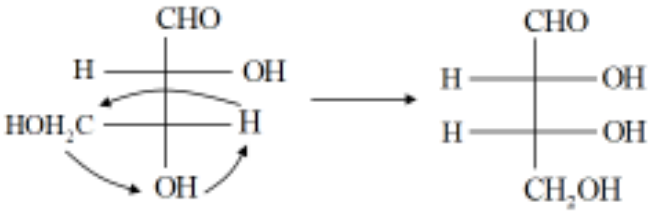
87) $\text{BP} \propto \text{Molecular weight}$

$\propto \frac{1}{\text{Branching}}$

88)

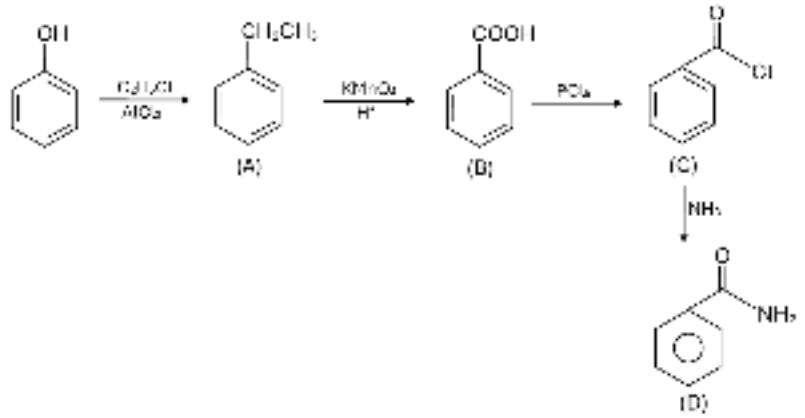


3-Bromo-5-methoxy benzaldehyde



89)
Both –OH ON
 Same side → Erythrose
–OH ON last
Chiral is ON right → D

90)



Hence, Option (2) is correct

BIOLOGY

91)

NCERT XI, Pg. # 32

92) NCERT Class-XI : Page No. 7, 18, 29, 30, 32

93)

NCERT-XI, Pg. # 32, 33, 40
Chara, Spirogyra, Volvox

94) NCERT Pg. # 11, 12, 26

95)

NCERT XI, Pg. # 28

96)

NCERT XI, Pg. # 64

97) NCERT Pg. # 58

98) NCERT XI, Pg. # 75,76

99) NCERT XI, Pg. # 73

100) NCERT XI, Pg. # 126

101) NCERT XI, Pg. # 101

102) NCERT, Pg # 99,98,96,95

103)

NCERT XI, Pg. # 123

104)

NCERT XI, Pg. # 93,94

105) NCERT-XI Pg # 118

106) NCERT Pg. # 107

107) NCERT, Pg. # 109, 110

108) NCERT Pg. 175,176,177

109) NCERT, Pg. # 146

110) NCERT, Pg. # 145

111) NCERT XI, Pg. # 154

112) NCERT-XI Pg # 160

113)

NCERT XI, Pg # 156

114) NCERT Pg. # 4

115)

NCERT XII, Pg. # 35

116)

NCERT XII, Pg. # 35

117) NCERT-XII Pg-63

118) NCERT-XII Pg # 69

119) NCERT Pg. No. # 69, 70

120)

NCERT XI, Pg # 89, 97, 101, 114

121)

NCERT XII, Pg. # 90

122) NCERT XII, Pg. # 97

123)

NCERT XII, Pg. # 93,94

124) NCERT XII, Pg. # 103

125)

NCERT XII, Pg. # 97,102,103

126) NCERT XII, Pg. # 75

127)

NCERT XII, Pg. # 106

128)

NCERT XII, Pg. # 198

129)

NCERT XII, Pg. # 197

130) NCERT Page No.: 245

131)

NCERT XII, Pg. # 211

132) NCERT Pg.# 243

133)

NCERT XII, Pg. # 222

134)

NCERT XII, Pg. # 220

135)

NCERT XII, Pg. # 223

136)

NCERT Pg # 49

137)

NCERT-XI, Pg. # 57,59

138) NCERT Pg. # 47

139)

NCERT Pg # 40

140) NCERT, Pg # 40, 41, 42, 43

141) NCERT Pg. # 102

142)

Module (4), Pg # 123

143) NCERT Page No. 120

144) NCERT-XII, Pg. # 117

145)

NCERT XI, Pg # 197

146) NCERT Page No. 201

147) NCERT Pg.#197.

148) NCERT Pg.#194-195.

149) NCERT-XII, Pg. # 287, 288

150) NCERT, Pg. # 189

151) NCERT, Pg. # 186-187

152) NCERT Pg. No. # 190

153) NCERT, Pg. # 209

154) NCERT (XI) ; Page No. # 207

155) NCERT XI Pg. # 241,242,243,246

156) NCERT Pg.# 231, 232.

157)

NCERT Pg # 236

158) NCERT Pg.#243.

159) NCERT Pg.#247.

160) NCERT Pg. # 248

161) NCERT XI Pg.#218

162) NCERT Pg. # 227

163) NCERT, Pg # 34, 35

164) NCERT(XII) Page#48/54(H) Para:3.5

165) NCERT(XII) Page#37/41(H) Para:2.6

166)

NCERT Pg#31,34

167) NCERT (XII) ; Page No. # 28, 29, 30

168) NCERT Pg. # 31

169) NCERT XII, Pg. # 137

170) NCERT XII, Pg. # 133

171) NCERT Pg. # 142, 143

172) NCERT Pg. #142, 143

173) NCERT BASED

174) NCERT XII Pg. No. # 125

175)

NCERT XII, Pg # 169
Option 3: A plasmid used to transfer DNA into a living cell.
In recombinant DNA, a vector is a vehicle (like a plasmid) that carries foreign DNA into a host cell for replication or expression.

176)

NCERT, Pg. # 165

177) NCERT-XII, Pg. # 171, 172, 173, 174

178)

NCERT XII, Pg # 182
Question Explanation: Identify which peptides remain in mature insulin.

Concept: Proinsulin maturation

Solution: The correct answer is option 1

A. Proinsulin → A-chain + B-chain + C-peptide
During maturation, C-peptide is removed, leaving only A and B peptides, connected by disulfide bonds.

Final Answer: Option 1

179)

NCERT XII, Pg # 180, 184
Explain Question: The question asks to match Column I with Column II**Concept:** Biotechnological applications.
Solution:

List I	List II
1. α-1 antitrypsin	III. Emphysema.
2. Cry IAb	IV. Corn borer.
3. Cry IAc	I. Cotton bollworm.
4. Enzyme replacement therapy.	II. ADA deficiency.

So, the correct answer is: 3. A-III, B-IV, C-I, D-II
Final answer: Option 3

180)

NCERT XII, Pg # 156
Explain Question : The diagram shows a biogas plant. The question asks to correctly match the labeled parts (a, b, c) with their names.
Concept : This question is based on Inlet → Cow dung + water slurry enters.
Digester → Anaerobic chamber where methanogenic bacteria decompose organic matter.
Gas holder → Collects methane-rich biogas.

Outlet → Sludge (spent slurry) is removed for use as fertilizer.

Explanation :

A → outlet pipe for sludge → sludge.

B → large chamber → digester.

C → inlet pipe where dung and water enter → dung and water.

This matches:

A—sludge, B—digester, C—dung and water.

Final answer : 1